Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions

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Abstract

The matter hypothesis of Observer Patch Holography (OPH) identifies stable massive matter with localized topological defects of native overlap transport, whose excess maintenance energy is read in the same clock as propagation. A finite construction realizes protected transport-history sectors under a specified coherent memory law. The native two-load configuration complex has exchange group $\mathbb{Z}/2$; its ordered-cover lift preserves two sharp quantum sectors. Giving each occupied port an independent internal state makes the odd loop act as $-$SWAP, with exact CAR signs and complete source instruments. This construction protects a memory-carried charge but does not localize a physical particle core.

An exact basis map joins the retained fifteen-state matter module to a specified chiral action. Its calibrated charged Yukawas and mixing remain inputs. Complete CAR source factors implement finite spatial propagation, particle-changing Higgs operations and both coherent and separately flagged collision holds. A joint experiment uses one calibrated clock and retains all field sectors, native records and quantum references. It exhibits a measurable difference between the two collision policies.

The original three-dimensional Weyl/contact action also admits a positive-energy infinite-lattice ground at fixed spacing and certified energy-density brackets. Its finite grounded cube has a completely resolved 78-line local charged spectrum and exact state-dependent retarded memory. A complete native experiment prepares its invariant ground mixture and reads both charge branches with a four-bin phase clock, retaining every outcome and first failure. The infinite ground has an explicitly bounded non-Gaussian four-point correlation, an ordered Euclidean hierarchy with time-reflection positivity, nonzero interaction memory and a spatially local inverse response away from real frequencies. Exact internal-spin selection extends this spectrum to every vector in the complete finite ground space; a positive finite-resolution energy POVM supplies all readout probabilities with controlled errors. The finite-gap dictionary fails the charged-lepton hierarchy for every such ground preparation under every common positive unit factor. A bounded-contact refinement constructs a continuum field net with all eight Weyl valleys, positive energy and finite propagation. The contact force has a resolved composite CAR channel. A signed-energy sum rule excludes the stated CPT vacuum action on the original fields, ground and clock at positive limiting coupling. Particle-hole symmetry fixes the alternative clock $H-(U/2)N$ and a distinct centered ground. The finite cube has a unique half-filled ground with hidden spin zero, a certified gap to other charge sectors and faithful one-site states. An exact two-site control supplies symmetric charged spectra. Uniform spectral estimates show that

this centered composite force escapes every finite energy window under bounded-coupling refinement. Free-vacuum cyclicity and locality identify the entire continuum hierarchy with eight free Weyl valleys, all with $E = v|p|$ and zero rest mass. This trajectory therefore cannot supply the required interacting massive spectrum. A different justified trajectory, physical species, gauge and detector maps remain to be constructed. A specified one-dimensional coupling continuation gives a controlled massive relativistic branch, but the bare parameters do not select that continuation or a unique mass. The failed finite mass dictionaries and unprotected defect candidates retain their explicit negative-control domains. The remaining physical task is to realize localized defects, their interactions and measured spectrum in one chiral $3 + 1$ theory with the required massive sectors and gapless electromagnetic field.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/BASIS.md

calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.
1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.
2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.
3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.
4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A

declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

From repair topology to a field experiment. A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/COMPLETION_PROGRAM.md

Protected transport sectors. On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps x to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with Q, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies X to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0, 5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I - X)/2$ and $M_h = \sum_{x \in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x \in D_h} p_h(x) |f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively I and X. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step Z multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

Internal states and fermionic exchange. Give each occupied port an independent internal space V. The map

$$|\{x < y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b} |0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite V. The odd loop acts as $-$SWAP, giving -1 on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V\psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X \otimes I)E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through E_V. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension m enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3) \times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module: its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

One calibrated action and its observables. The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\mathrm{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\mathrm{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\mathrm{vac}}$ from the replicated vacuum scalar $12E_{\mathrm{vac}}$; the remaining common E_{vac} does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

Spatial dynamics, clock and particle creation. A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3+1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock source gates and fermionic routing determine their full Gaussian unitary maps without allocating arbitrary 2^{1080}-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1,0,0,1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If A is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1-A)^2$ for the flagged policy and one for the coherent

policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$, the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

A joint source experiment. Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814 before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is 4.200002×10^{-7}, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

Energy and modular observations on the same field. The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let H be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by e^{-itH}. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed H.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static H gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter s does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma\|\rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

The Weyl/contact branch and its finite premise closures. The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

A complete charged internal sector. The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can still steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

Complete extension and an exact spectral exclusion. Let P be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect P; adjoining the outside outcome $I - P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport q times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns V, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

A protected interacting sector on the original three-dimensional cube. The source-bound `cube_eta_protection` packet retains the original open $2\times2\times2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H, \eta^+] = U\eta^+$, checked exactly on the complete field action. For $X = \eta^+|_{N=4} : \mathcal{H}_4 \to \mathcal{H}_6$,

$$X^\dagger X = (\eta^-|_{N=4})^\dagger(\eta^-|_{N=4}) + 4I, \qquad H_6 X = X(H_4 + UI).$$

Consequently X is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4} = 1820$. Its range is precisely the $\eta \geq 2$ subspace at $N = 6$.

The executed repair domain is a finite four-block service: the four x_1-directed dimers, native ports $(0, 1, 5, 7)$, and all five inherited seams $(0, 1), (0, 5), (0, 7), (1, 5), (1, 7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N = 6$ support P. Its physical columns are expanded exactly as $A + \sqrt{17}B$, with integer Gaussian matrices A, B; $\eta^- A = \eta^- B = 0$, so the support has $\eta = 1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n = 136 - 32\sqrt{17}$, with $(A + \sqrt{17}B)^\dagger(A + \sqrt{17}B) = n^2 I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q = I - P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX = 0, \qquad L_r X = 0 \quad (r \text{ native}), \qquad QX = X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

Exact interacting spectral census and a local fixed-number probe. Let $v = \eta^+|0\rangle$, $v^\dagger v = 8$, and $\Pi_3 = I - vv^\dagger/8$ on the 120-dimensional $N = 2$ space. For $T = (\eta^+)^2|_{N=2}$,

$$T^\dagger T = 60I + 24vv^\dagger, \qquad H_6 T = T(H_2 + 2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta = 3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z = 16(E_{N=6} - 2)$,

$$p_{119}(z) = z^{57}(z^2 - 12)^{21} q(z) c_{64}(z)^3 c_{128}(z)^3,$$
$$q(z) = z^2 - 16z - 12, \qquad c_a(z) = z^3 - 16z^2 - 12z + a.$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies. Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N = 6$ reference and one-site density probe are

$$|R\rangle = \frac{(\eta^+)^3|0\rangle}{\sqrt{2016}}, \qquad H|R\rangle = 3|R\rangle, \qquad |s\rangle = \frac{4n_0 - 3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R = 3/4$, $\mathrm{Var}_R(n_0) = 15/16$; $|s\rangle$ is an $\eta = 3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0 - v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $1488395072912184839188800532906554156908544 > 0$. Together with the annihilator $qc_{64}c_{128}$ this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance 45/1792. This is a finite local unitary disturbance of an entangled reference, not yet a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled F_0, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)} F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified H and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

Native preparation and the remaining detector premise. The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by 8.48×10^{-13}, including every source export. All 130 following repair records are retained

without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification remain open. The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

An internal observer reads the protected response. The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[3]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$p_0(t,\phi) = \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\},$$
$$C(t) = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1.$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to H, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

Complete source and field error, without a sector projection. The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct

[3] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), https://arxiv.org/abs/1307.0006.

source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-H comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm $1/16$ and eight contact terms of norm one. Counting only possibly nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than 8.56×10^{-11} for the preparation, controlled pointer operations, and both complete source field words. The complete output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most ε, and each reconstructed quadrature by at most 2ε. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1, 0) < 0.994850, \qquad 0.501379 < p_0(1, \pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-H values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately 2.27×10^{-6}, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most $1/2048$ in these two probabilities using 2^{24} trials per setting, except with probability less than $4(3/8)^8$.[4] The sampling allowance still separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of 2^{24} trials; the complementary count includes every other complete outcome.

[4] The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, https://doi.org/10.1080/01621459.1963.10500830.

What eight-line resolution would additionally require. The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \dots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k - \omega_j)}{2}\right| > 1400.$$

Hadamard's inequality bounds every 7×7 cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w} - w\|_\infty < \frac{43904}{2625}\, e,$$

where e bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error $1/65536$ and sampling error $1/32768$ at each of the sixteen settings. With $9 \cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than $343/448000$. Twice that error is smaller than the least certified weight, so every line is separated from zero. The total is 77 309 411 328 trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

Full weight resolution by certified accelerated emulation. The preceding sixteen-setting proposal can now be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight $1/16$.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical η^+. The complete contact layer satisfies $D_6T = TD_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h)T = e^{-2ih}TS_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If U is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD)W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication. For approximate unitaries with errors a, b, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by 1.323294×10^{-12}. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix B for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,$$
$$.004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

Comparing the actual interacting regional clocks. The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

The stronger identification fails on this reference. For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed 1/8, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than 10^{-4}, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

Controlled regional modular energy. The same complete-Fock Weyl/contact action now supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace E, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below 1.917×10^{-9}. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than 2.7×10^{-8} from $\beta C_1 + aI$ for every scalar a on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the

norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

A separately prepared interacting Gibbs reference. On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below 1.700×10^{-12} on the initialized input. The actual regional success logarithm, including the exact normalization scalar, is within 2×10^{-9} of the preceding P_8 comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

The actual stationary source weight. One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \operatorname{diag}(a,b)^{\otimes M} = a^M e^{\gamma N}$, where $a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below 1.894×10^{-16}, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a k-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2+\epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from βK, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are in `model/continuing_reference/README.md`; the following construction closes its previously restricted scheduler premise.

A stationary finite controller for native feedback. The reference can now continue under an explicit record-adaptive scheduler. Use two classical working bits (c, d); $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record r, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \,\mathrm{XOR}\, f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output (u, v) every incoming term uses the same instrument v. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \operatorname{diag}(a,b)^{\otimes M}.$$

Here a, b are the actual literal-source weights, so the successful block still has trace $(a+b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all

6^M source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must still be added.

A complete finite mixing certificate and its boundary. On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is $\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the 12^2 input/output population pairs, with common probability denominator 2^8. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3,1,2,1,3,2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

Exact energy Ward cells under coarse-graining. The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all 2^{54} Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} \big(I_{x,i}(n) - I_{x-e_i,i}(n)\big) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n,x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n,x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\mathrm{coarse}} P_{\mathrm{faces}} = P_{\mathrm{sites}} D_{\mathrm{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U + 3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If L_h, L_F are the existing normalized locality

tails and ϵ is the patch unitary error, an endpoint energy error is at most $A\min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB\min(2, L_F + 2\epsilon)$. Replacing the integral by m midpoint samples adds the explicit bound $t^2 B\, 343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full Einstein premise register.

Local momentum balance retains its force. On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the 3^3 box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

Global periodic momentum and the local question. On a declared periodic box, graded translations T_j commute with the full interacting action. Their principal Hermitian logarithms P_j are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-R density sum obeys $\|P_j - A_R\| \geq 1/(R+1)$ when $L > 2(R+1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

A collision-suppressed local Ward bound. The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let B be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let Ω restrict the initial state to that band and $N \leq N_{\max}$. Write $D = \sum_x n_{x\uparrow} n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D \Omega \leq f \min(N_\uparrow, N_\downarrow) \leq fd$. Since $D^2 \leq dD$ on this number sector,

$$\|UD\Omega\| \leq Ud\sqrt{f}, \qquad \|(I - \Omega)e^{-itH}\Omega\| \leq \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of Ω and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K - B)|/V$, giving escape norm squared $\max_K \lambda_K(1 - \lambda_K)$ before multiplication by U^2. Thus projecting the outgoing contact back into B would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) = i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \leq U$ and $\|F_j(x)\Omega\| \leq 4Uf$. For preparation at zero and $0 \leq t_0 \leq t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[s_j(x, t_1) - s_j(x, t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\leq \min\left\{ U(t_1 - t_0),\ 4Uf(t_1 - t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode 3^3 source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \le \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \le 2\kappa$ even after relative-momentum scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \le \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a 64^3 counting example with $B = \{-1, 0, 1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after division by κ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

Interacting lattice scattering and its zero threshold. The same Weyl/contact action now admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum K, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm U. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[5] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n \ge 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \le 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree M, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for π give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering

[5] The imported trace-class and finite-rank scattering results are reviewed in https://arxiv.org/abs/1711.00528 and https://arxiv.org/abs/0902.3568; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

Interacting edge scattering and a moving bound pair. The same two-body source family now has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3 \gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

The normalized full-torus integral satisfies $2/3 \le W_0 < 5$; $a\gamma \le 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither γ nor this clock/energy choice is inferred from physical data.

At zero total momentum the threshold estimate is

$$\left| \frac{g_0(6/a^2 + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le 19a|z|$$

for real $|a^2 z| \le 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the respective Green remainders 18η and $36|\eta|$ before rescaling. Thus, for $0 \le s \le S$, $a^2 S \le 1/8$ and $836aS \le 7\gamma$,

$$S_J(s) \longrightarrow \frac{\gamma - i\sqrt{s}}{\gamma + i\sqrt{s}}, \qquad \sup_{[0,S]} |S_J - S_{J,\lim}| \le \frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

Nonzero total momentum has a controlled pole. Write $K = aP$, $q = k - K/2$, $d_j = \sin q_j \cos(K_j/2)$, $e_j = \cos q_j \sin(K_j/2)$ and $R^2 = \sum_j \cos^2(K_j/2)$. For $|K_j| \le \pi/2$, concavity gives the exact complete free edge $B_a(K) = 2c_a R$, attained at all eight $q_j = \pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q, E) = \frac{E(E^2 - 4c_a^2 e^2)}{E^4 - 4c_a^2(d^2 + e^2)E^2 + 16c_a^4 (d \cdot e)^2}.$$

At $E = B_a(K) + z$, put $t = E/(2c_a)$, $\eta = t^2 - R^2$ and $V = \sum_j \cos K_j \cos^2 q_j$. Equivalently,

$$g_K(E) = \frac{t}{2c_a} \left\langle \frac{1 - e^2/t^2}{V + \eta + (d \cdot e)^2/t^2} \right\rangle.$$

Let $\kappa = \max |K_j| \le 1/2$ and $h = a^2 z \le 1/8$, with $z \ge 0$. Writing $Q = \sum_j \cos^2 q_j$, the denominator lies between $(1 - \kappa^2/2)(Q + h + h^2/12)$ and $Q + h + h^2/12$; $e^2 \le \kappa^2 Q/4$ and $(d \cdot e)^2/t^2 \le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K) + z) - g_0(6/a^2 + z)| \le 3a^2 \kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left| \frac{g_K(B_a(K) + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le a(11z + 3P_{\max}^2), \qquad |P_j| \le P_{\max}.$$

Under $256a(\gamma^2 + P_{\max}^2) \le \gamma$, the pole denominator has opposite signs at $z = \gamma^2/4$ and $z = 4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K) = B_a(K) + \delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4} \le \delta_a(K) \le 4\gamma^2, \qquad |\delta_a(K) - \gamma^2| \le \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U > B_a$.

An explicit bound-band wavepacket comparison. The same global J_a has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing B_a and bounding the sine remainder gives

$$\sup_{|P_j| \le P_{\max}} \left| \epsilon_a(P) - \left(\frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d / \langle d, R_K^2 d \rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K, k) = a^{-3/2} f(K/a) \psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If h_{pair} multiplies by $|P|^2/4 - \gamma^2$, then for arbitrary untouched references,

$$\| e^{-itJ_a} W_a - W_a e^{-ith_{\text{pair}}} \| \le \min\{2, |t| \mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-K on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S - I|^2 dE \le 1024 M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \le c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

Identified model energies and the physical particle boundary. The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy γ^2 and inertial mass 2 in those model units. If physical energy and length units $E_{\text{u}}, L_{\text{u}}$ are independently attached and $p_{\text{phys}} = \hbar P / L_{\text{u}}$, then $M = 2\hbar^2/(E_{\text{u}} L_{\text{u}}^2)$ and $E_{\text{bind}} = E_{\text{u}} \gamma^2$, with $E_{\text{u}} T_{\text{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

H energy	$E-3$	Probe weight
1.816498920	−1.183501080	0.002465366
1.860537501	−1.139462499	0.005644624
1.955137632	−1.044862368	0.005881219
2.108692846	−0.891307154	0.007137664
2.167639734	−0.832360266	0.004528123
3.015861346	+0.015861346	0.421577939
3.030769653	+0.030769653	0.415789141
3.044862368	+0.044862368	0.136975924

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N=6$ ground: $-3\sqrt{3}/8 \le E_{0,N=6} \le -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N=6$: the transition operator is number-neutral and even under $\Gamma=(-1)^N$. An isometric dictionary intertwining this grading cannot map that response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[6]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own H. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N=6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy. These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal still leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

[6] https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf.

The preceding identity audit leaves a relativistic construction open. The following extension constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

A controlled massless relativistic sector of the unchanged action. The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\leq N}$ on all particle numbers through a fixed N, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1 s_2 s_3$, the limiting blocks are

$$h_b(p) = -v\sum_j s_j \sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity β, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless Weyl/Dirac in coordinate $x^0 = vt$, not merely a fitted linear dispersion.[7] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

The full interacting evolution has an explicit comparison bound. The sine remainder gives, uniformly on C,

$$\|h_a(\nu + ap) - h_b(p)\| \leq \frac{va^2}{6}\sum_j |p_j|^3 \leq \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \leq 1$ and $p \neq 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2\sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let Ω_a project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining D are bounded by d, hence converge strongly and satisfy $D^2 \leq dD$. On band states each annihilation has

[7] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html.

norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms gives $\Omega_a D\Omega_a \le f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \le d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula now yields

$$\|e^{-itH_a}W_a^{\le N} - W_a^{\le N}e^{-itd\Gamma(h_{\mathrm{W}})}\| \le \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it bounds half trace distance and individual probability errors, capped by one. For fixed N, Λ, v and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \le 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

The finite source check retains all 128 one-particle Fourier columns on a periodic 4^3 cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1-\Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

Resulting energies and the physical boundary. At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

p	lattice $\lvert E_a\rvert$	continuum $\lvert E\rvert$	continuum mass
$(0,0,0)$	0	0	0
$(1,0,0)$	0.999959310393	1	0
$(1,1,0)$	1.414156018578	$\sqrt{2}$	0
$(1,1,1)$	1.731980331101	$\sqrt{3}$	0

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: already at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every U. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single 2×2 constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

A detector identifies the model's propagating Weyl flavors. The massless sector now has a concrete model-particle readout. On the periodic 8^3 lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi\ell_b/4$. For component eigenvalue $r = \pm 1$ of σ_z, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b\cdot x}}{\sqrt{512}}\delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \geq 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A+\sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with H. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

flavor b	momentum labels ℓ_b	positive-branch component	positive cone helicity
000	$(0,0,1)$	1	$-1/2$
001	$(0,0,5)$	0	$+1/2$
010	$(0,4,1)$	1	$+1/2$
011	$(0,4,5)$	0	$-1/2$
100	$(4,0,1)$	1	$+1/2$
101	$(4,0,5)$	0	$-1/2$
110	$(4,4,1)$	1	$-1/2$
111	$(4,4,5)$	0	$+1/2$

Here helicity uses the explicit proper valley frame $R_b = -\chi_b\,\mathrm{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y}r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c\cos k_z\,\sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

The Fourier readout is compiled with complete native errors. The detector applies the fermionic extension of $F[k,x] = e^{-2\pi ik\cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4 + 1)$ proves the 64 entries and 64 Gram sums of the one-axis

transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-H propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-H propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most ϵ_{source}.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about 1.94×10^{-15}, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture. These are model-simulated counts, not a literal 2^{1024}-row native execution or laboratory data. No survival normalization is used.

Mass readout and comparison with measured particles. For the periodic family $L = 8 \cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_{\pm}(a) = \pm \frac{\sin(a\pi/4)}{8a} \longrightarrow \pm \frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

physical target	measured mass in MeV/c^2	present continuum mass
electron	$0.51099895000 \pm 0.00000000015$	0
muon	$105.6583755 \pm 0.0000023$	0
tau	1776.93 ± 0.09	0

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[8] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[9] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are now operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained

[8] PDG 2025 listings: https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf.

[9] The photon listing is https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported 3σ fit ranges, not exact bounds: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf.

rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

A global interacting ground on the original cube. The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, now admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim\ker(H - E_G) = 5.$$

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state G. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix Q; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40}QQ^T - \frac{27}{64}I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since QQ^T is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above E_G. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated H.

Positive charged spectral data above this ground. At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

probe or component	N	weight	$\omega = E - E_G$
$c_\uparrow G$	3	1/2	λ
$c_\uparrow^\dagger G$	5	1/2	λ
$c_\downarrow G$	3	0	absent
$S = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	λ
$\eta = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	$1 + \lambda$

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is 4/5, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_\downarrow^\dagger G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5} S^- S^+ v, \qquad v_\eta = \frac{1}{5} \eta^+ \eta^- v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at -3λ and $1 - 3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_\downarrow = \frac{1}{10}\delta_\lambda + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp} \mu_r \subset (7/1024, \infty).$$

The displayed 1/10 values are exact symmetry-resolved contributions; without further resolution of μ_r, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \operatorname{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \operatorname{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in v is 1/2; deleting that interaction changes its mean gap from 1/2 to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and -1 and odd CAR grading relative to G; electric charge and physical Lorentz spin do not follow from these labels.

Attachment to the declared repair process. The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original H preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for G, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

An interaction-generated charge scale in the original one-dimensional law. There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L, 1, 1)$, $L = 4m$, the onsite frame $G_x = (-1)^x (i\sigma_x)^x$ satisfies $G_x(i\sigma_x)G_{x+1}^\dagger = -I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t = c/2 = 1/16$ and $U = 1$. Four exact local frame identities, source-matrix checks through $L = 16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c = \lim_{L\to\infty} [E_L(L+1) + E_L(L-1) - 2E_L(L)] = U - 4t + 8t \sum_{k\geq 1} \binom{1/2}{k} (2t/U)^{2k-1} \eta(2k-1).$$

Here the series is used only for $2t/U < 1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c = 0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail 2^{-96}. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original H:

$$E_+ = \frac{U + \Delta_c}{2} = 0.885775857869\ldots, \qquad E_- = \frac{\Delta_c - U}{2} = -0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu = 0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2 = 0.3857758578690188\ldots$ uses the explicitly centered $K = H - (U/2)N$; Δ_c itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact still vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

The fixed-interaction continuum boundary. The equivalent positive integral for the charge gap and the elementary K_1 integral bounds imply, for every $t, U > 0$,

$$\Delta_0 \leq \Delta_c \leq \Delta_0 \frac{1 + 3/(8A)}{1 - e^{-2A}}, \qquad A = \frac{2\pi t}{U}, \qquad \Delta_0 = \frac{8}{\pi}\sqrt{Ut}\, e^{-A}.$$

Consequently fixed U with $t = v/(2a)$ has $\Delta_c \to 0$ as $a \to 0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A) = \frac{4ve^{-A}}{M\sqrt{2\pi A}}, \qquad t(A) = \frac{v}{2a(A)}, \qquad U(A) = \frac{\pi v}{a(A)A},$$

for a *chosen* $M, v > 0$. It gives $\Delta_0/2 = M$ exactly and $1 \leq \Delta_c/(2M) \leq [1 + 3/(8A)]/[1 - e^{-2A}] \to 1$. This is a quantitative charge-threshold limit, not a derivation of M from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

A controlled relativistic charge mass on an explicit running family. The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here p is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi} \sum_{\substack{m \geq 1 \\ m \ mod d}} \frac{K_1(mA)}{m} \cosh(m\theta), \qquad vp = \frac{8t}{\pi} \sum_{\substack{m \geq 1 \\ m \ mod d}} \frac{K_0(mA)}{m} \sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly M. For $A \geq 1$ and $|\theta| \leq \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1-q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1 - 1/(8z) \leq K_0(z)/b(z) \leq 1$ and $1 \leq K_1(z)/b(z) \leq 1 + 3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \leq r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \leq r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \leq \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \longrightarrow 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy M. The initial lattice half-gap is different, and the fixed-U continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

The original three-dimensional action also permits a growing-volume ground construction at fixed spacing. This closes ground-state existence and positive excitation energy without selecting the one-dimensional canonical reference or its coupling continuation. Its local response is then connected to the existing geometry theorem interfaces.

An original-action infinite-volume ground representation at fixed spacing. The full three-dimensional Weyl/contact action now admits a translation-invariant, gauge-invariant ground state at its original zero chemical potential, without choosing a particle-number sector. This is a compactness existence theorem, not a computed or natively prepared state. Keep fixed finite $c > 0$, $U > 0$ and the complete quasi-local CAR algebra with two modes per site. The previously proved full-Fock locality estimate supplies the strongly continuous dynamics τ_t. On each even-sided cube Λ_L of side $2L$, take the normalized projector onto the absolute minimum of H_{Λ_L}, extend this even state by the exterior vacuum, and average its translates over all sites of Λ_L. A diagonal subsequence of the bounded local matrix entries defines a state ω; positivity, normalization and CAR compatibility pass to the limit. The vanishing boundary-to-volume ratio makes ω translation invariant.

For a local A, including an odd charged probe, the local derivation $[H, A]$ contains only interaction terms touching its support. Whenever the support and its nearest neighbors lie inside the box,

$$\omega_L(A^*[H, A]) = \operatorname{Tr} \rho_L A^*(H_{\Lambda_L} - E_L)A \geq 0.$$

Bad translates occupy a vanishing fraction. Their real-part lower defect and imaginary magnitude are each bounded by $2\|A\|^2|\operatorname{supp} A|(U + 6c)$, so $\omega(A^*[H, A]) \geq 0$ in the limit. This uses the full ground minimum and would fail for a merely canonical ground with an unrestricted charged probe.

For a radius-R buffer, let $B_R(t)$ denote the existing volume-uniform local-dynamics boundary bound. Comparing both finite and infinite dynamics to this intermediate cube gives, for $\|A\| \leq 1$,

$$\left|\nu_L(\tau_t(A)) - \nu_L(A)\right| \leq \min\Big\{2,\, 2B_R(t) + 2\left[1 - (1 - R/L)^3\right]\Big\}.$$

First $L \to \infty$, then $R \to \infty$, establishes stationarity. The same comparison applies to two-point correlations. Every finite full-ground correlation has a positive spectral measure supported on $[0, \infty)$; compact-time convergence and bounded correlations pass all Schwartz Fourier tests supported on negative energies to the limit. The GNS implementation therefore satisfies

$$U_t\pi(A)\Omega = \pi(\tau_t(A))\Omega, \qquad U_t = e^{itL_\omega}, \qquad L_\omega \geq 0, \quad L_\omega\Omega = 0.$$

The generator measures excitation energy in this representation; no chemical potential has been inserted into the source action.

The state can be chosen at nonzero density. Tiling each even box by the already certified four-particle cube vector gives contact zero, energy $-2\sqrt{3}c$ per eight-site tile, and zero expectation for every cross-tile hopping. Hence $E_L/|\Lambda_L| \leq -\sqrt{3}c/4$. Cartesian difference operators commute and the Pauli mixed terms cancel, giving $h^2 \leq 3c^2 I$ and $H \geq -\sqrt{3}cN$. Thus every such full ground has density at least $1/4$. The exact real-bipartite particle–hole identity $E(2V - N) = E(N) + U(V - N)$ excludes $N > V$ from an absolute minimum when $U > 0$. Spatial averaging before the subsequence passes both bounds:

$$\frac{1}{4} \leq \omega(n_{0\uparrow} + n_{0\downarrow}) \leq 1.$$

Half-filling and a particular density are not assumed or concluded. The original ground energy per site also has a limit e_∞. For every even cube side ℓ, tiling larger boxes gives the computable bracket, conditional on its finite full-ground energy,

$$\frac{E_\ell}{\ell^3} - \frac{3c}{\ell} \leq e_\infty \leq \frac{E_\ell}{\ell^3}, \qquad -\sqrt{3}c \leq e_\infty \leq -\frac{\sqrt{3}c}{4}.$$

The lower surface term counts intertile hopping norms; a product of gauge-invariant tile grounds gives the upper bound. Untiled boundary terms vanish per volume, and the bracket width tends

to zero. For the constructed averaged ground, $\omega(h_0) = e_\infty$, where h_0 assigns half of each incident bond to the origin. This supplies an energy-density observable, not a state-convergence rate or rest mass.

The contact interaction remains represented:

$$[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3) \neq 0.$$

Every unital GNS representation is injective on each full local matrix algebra, so this quartic operator cannot disappear in representation. This faithfulness of the representation does not assert faithfulness of the ground-state functional. The claim is a nonzero represented interaction derivation, not a scattering or connected-vacuum-correlator theorem.

An exact source countercontrol prevents simply substituting the free filled sea. For a pure Slater projector with $P_{xx} = I_2/2$,

$$\mathrm{Var}(D) = \frac{V}{16} + \sum_{x \neq y} |\det P_{xy}|^2 \geq \frac{V}{16}.$$

If it is also a kinetic eigenstate, its interacting energy variance is $U^2 \mathrm{Var}(D) > 0$. On the original cube, the complete 3844-term negative-band filling has $N = 8$, $\langle D \rangle = 2$ and $\mathrm{Var}(H) = 2/3$ at $U = 1$. The exact checks and proof are retained in `model/thermodynamic_ground/results.json`. For local CAR probes, the same subsequence also supplies the analytic retarded response

$$G_{AB}(z) = -i \int_0^\infty e^{izt} \omega(\{\tau_t(A), B^*\})\, dt, \qquad \mathrm{Im}\, z > 0.$$

Each time tail after T is at most $2\|A\|\|B\| e^{-\eta T}/\eta$ for $\mathrm{Im}\, z \geq \eta > 0$. Compact-time locality and weak-star convergence, followed by this tail bound, give uniform convergence on compact subsets of the upper half-plane. The unknown finite-state convergence error is retained; no real-axis pole, local inverse or metric is inferred. The result closes fixed-spacing ground-state existence and the positive-energy representation for the actual interacting lattice. It supplies no selected ground functional, convergence rate, native preparation, repair stationarity, uniform gap, Lorentz continuum, physical particle assignment or Einstein dynamics.

Exact matter response before a geometric identification. The unchanged open $2 \times 2 \times 2$ Weyl/contact cube at $c = 1/8$, $U = 1$ now supplies a state-dependent retarded response, including its exact eliminated-state memory. In the existing onsite hidden-spin frame, write $h = A/16$, where A is the real signed cube adjacency and $A^2 = 3I$. The retained full-Fock ground theorem gives the normalized highest-weight reference G, with four occupied negative-band up orbitals, $E_G = -\sqrt{3}/4$, and a strict gap $\delta = 7/1024$ to every different particle-number sector. The second exact stationary reference is $G_+ = \sqrt{2}\, c_{0\uparrow}^\dagger G$, of energy $-3\sqrt{3}/16$. No chemical potential, fitted mass, or new geometry law is inserted. Physical CAR responses are obtained by the existing onsite unitary frame; hidden spin is not assigned a physical Lorentz or electric-charge meaning.

For either reference ψ of energy E, the complete eight-site fermionic response, for $\mathrm{Im}\, z > 0$, is

$$\begin{aligned} \mathcal{G}_{xy}(z) &= -i \int_0^\infty e^{izt} \langle \{c_x(t), c_y^\dagger\} \rangle_\psi \, dt \\ &= \langle c_x [z - (H - E)]^{-1} c_y^\dagger \rangle_\psi + \langle c_y^\dagger [z + (H - E)]^{-1} c_x \rangle_\psi. \end{aligned}$$

There is an exact isometric dilation

$$\begin{aligned} V_y &= c_y^\dagger \psi \oplus \overline{c_y \psi}, & V^\dagger V &= I, \\ L &= (H_{N+1} - E) \oplus -\overline{(H_{N-1} - E)}, & \mathcal{G}(z) &= V^\dagger (z - L)^{-1} V. \end{aligned}$$

The conjugate removal space fixes the matrix indices; the present source arrays happen to be real. With $P = VV^\dagger$, $Q = I - P$, define $M_1 = V^\dagger LV$, $W = QLV$ and $B = QLQ|_{\operatorname{ran} Q}$. Exact block elimination gives

$$\mathcal{G}(z) = \left[zI - M_1 - \Sigma(z)\right]^{-1}, \qquad \Sigma(z) = W^\dagger (z - B)^{-1} W, \qquad W^\dagger W = M_2 - M_1^2.$$

Both L and B are self-adjoint; consequently $-\operatorname{Im} \mathcal{G}(z) > 0$ and $-\operatorname{Im} \Sigma(z) \geq 0$ in the upper half-plane. The eight observed columns do not replace the source Hilbert space: their exact complement remains in the memory kernel.

A computed change of response at fixed microscopic geometry. For the up channel of both references, the full cyclic identity $LV = Vh$ gives $\mathcal{G}_\uparrow(z) = (z - h)^{-1}$ and $W = 0$. For G, the up dilation retains all 56 addition and 56 removal states. The ground down channel has no removal component and retains all 560 states with $N_\uparrow = 4$, $N_\downarrow = 1$. Exact arithmetic over $\mathbb{Q}(\sqrt{3})$ gives

$$M_{1,\downarrow} = h + \tfrac{1}{2} I, \qquad W_\downarrow^\dagger W_\downarrow = \tfrac{1}{4} I.$$

Thus a static Hartree term alone does not reproduce this interacting response. Indeed $W_x = (n_{x\uparrow} - 1/2) c_{x\downarrow}^\dagger G$; orthogonal down locations and up occupation variance $1/4$ give the displayed Gram matrix directly.

For G_+, the down channel retains its entire 448-state invariant block. In lexicographic order of the eight cube sites, its exact matrices are

$$M_{1,\downarrow}^+ - h = \operatorname{diag}(1, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$
$$(W_\downarrow^+)^\dagger W_\downarrow^+ = \operatorname{diag}(0, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The first-moment change is exactly the added orbital density, $(1/2, 1/6, 1/6, 0, 1/6, 0, 0, 0)$. Removing the contact term in the explicit $U = 0$ control removes both this Hartree term and the ground down memory, giving $(z - h)^{-1}$. That comparison does not declare G a full free-Fock ground. The excited reference receives no positive-gap assertion.

The complete local charged spectral measure. For the original physical origin probe $v = c_{0\downarrow}^\dagger G$, the retained calculation now resolves all 78 distinct positive spectral lines, including the previously unresolved weight 4/5. A monic degree-78 polynomial p over $\mathbb{Q}(\sqrt{3})$ is admitted by the exact source equation $p(16H)v = 0$. A nonzero 78×78 Hankel determinant modulo 97, with $\sqrt{3} \mapsto 10$, proves the matching cyclic rank lower bound. Exact moments determine a numerator q such that

$$\langle v, (z - 16H)^{-1} v \rangle = \frac{q(z)}{1296\, p(z)}.$$

The denominator 1296 is the squared norm of the inherited integer state numerator. Modular reconstruction and a numerical eigensolver only propose coefficients and brackets. Integer annihilation and rank checks, 78 disjoint strict sign changes, and outward 1024-bit interval arithmetic certify every root and its strictly positive residue $q(x)/(1296 p'(x))$. Each charged gap is $(x + 4\sqrt{3})/16$.

The smallest gap is approximately 0.029950938342053226, with weight 0.3130113772155167; the largest is approximately 1.8360744654423855, with weight $2.711810625844545 \times 10^{-10}$. The receipt retains every gap and weight in rational intervals rounded outward to 24 decimal places. Polynomial substitution proves that the two lines $\sqrt{3}/16$ and $1 + \sqrt{3}/16$ each have total weight exactly 1/10. The remaining 76 lines have total weight 4/5, leaving zero unresolved weight for this local probe. This is the complete local cyclic measure, not the entire 560-state spectrum, a native detector execution, an infinite-volume pole theorem, or a relativistic mass assignment.

Controlled low-frequency coefficients. For the interacting ground down channel, put $K = H_{4,1} - E_G$. The full-ground certificate gives $K > \delta I$. Moreover, $D \le 1$ in this block, the kinetic norm is at most 5λ, and $-E_G = 4\lambda$, with $\lambda = \sqrt{3}/16$; hence $K \le 1 + 9\lambda < 2$. The same bounds hold for its compression B. For $|z| \le r < \delta$ the exact memory has the convergent expansion

$$\Sigma(z) = -\sum_{n=0}^{p} z^n C_n + R_p(z), \qquad C_n = W^\dagger B^{-n-1} W \ge 0,$$
$$\|R_p(z)\| \le \frac{r^{p+1}}{4\delta^{p+1}(\delta - r)}.$$

These are finite gapped frequency coefficients, not a spatial derivative expansion or a relativistic cone. All retarded moments $M_0, \dots, M_4$ are also computed exactly; the corresponding large-frequency remainder is $2^{p+1}/[R^{p+1}(R-2)]$ for $|z| \ge R > 2$.

The static response is computed with an exact residual certificate. For $C = V^\dagger K^{-1} V = -\mathcal{G}_\downarrow(0)$,

$$\|C - (12.9058381883 I - 4.9026412320 A)\| < 10^{-7},$$
$$\|\Sigma(0) - (-0.3633623064 I - 0.0105943755 A)\| < 10^{-6}.$$

A numerical solve only proposes a matrix with denominator 2^{40}. Integer arithmetic encloses its full $\mathbb{Q}(\sqrt{3})$ residual; division by the proved δ bounds the solution error. The exact isometry then bounds C, and $C > I/2$ bounds its inverse. An independent rational inverse residual certifies $\Sigma(0) = C^{-1} - M_1$. No numerical tolerance admits the theorem, and exact equality to an affine adjacency polynomial is not inferred.

The existing geometry interface and its remaining join. The same-action thermodynamic-ground construction separately supplies analytic local responses along its ground subsequence on compact subsets of $\operatorname{Im} z > 0$. The cube calculation does not supply a rate of convergence to that state, and its finite gap is not transferred to infinite volume. The existing `wave_density_metric` constructor takes a complete local action density $\mathcal{C} = \mu \operatorname{diag}(-w, a)$, with $\mu, w > 0$ and $a > 0$, and recovers $g = \sqrt{-\det \mathcal{C}}\, \mathcal{C}^{-1}$. The existing `source_tetrad` interface in the common matter join then constructs the common coframe and the scalar, Weyl, Dirac and Maxwell principal-symbol identities for its supplied density. The present result supplies an actual matter-dependent inverse response from which such coefficients must be derived. It does not substitute a Hartree diagonal for a spatial principal symbol, identify the cube with the distinct screen source, or complete the common action-normalization join. Fixed microscopic geometry is compatible with this emergent-response route; a new fundamental metric register is not required. Common local limits, modular and area response, full stress, and the same-family Einstein balance premises remain to be proved. No new physical species, native response detector, or completed G1–G6 premise is claimed.

The finite ground and its charged probe can now be prepared and read by the declared observer service. The full matrix spectrum extends the local probe calculation. Separately, the infinite ground admits a spatially local inverse response with nonzero interaction memory at complex frequencies separated from the real axis. These constructions supply additional inputs to the existing field and geometry interfaces.

Native preparation of the same ground and its charged response. The original cube ground and its local charged spectral measure now have an explicit native source preparation and readout. The exact highest-weight ground G, with $N = 4$ and $E_G = -\sqrt{3}/4$, has 62 nonzero hidden-up occupation amplitudes $(a + b\sqrt{3})/36$, where a, b are integers and $ab = 0$. Its squared amplitudes

are therefore exact integers divided by 1296. These source-derived weights and signs determine an eight-bit binary amplitude tree; no fitted spectrum or unknown-state oracle is used. The existing compiler emits 5,379 literal native pulses. An independent rational interval check consumes every event, inverts the Walsh layers, and verifies the sine and cosine of each conditional rotation against the exact signed square-root ratios. Disjoint control blocks give a maximum error per level, and telescoping the levels and complete native load/export maps bounds the full initialized preparation, including aborts.

Eight blank down modes are included by an exact isometric embedding into all 65,536 Fock rows. A further 184 native pulses apply the inverse onsite frame. Each local frame word is checked on its full four-state Fock space, including vacuum and doublon phases, using $e^{is\pi P/4} = (I + isP)/\sqrt{2}$ and Gaussian-integer products. Its literal-angle and native-instrument errors are separately enclosed. The resulting complete preparation-state error is below 10^{-10}; the retained exact bound is approximately 8.67×10^{-13}.

At the origin the hidden and original frames coincide, and the down mode is empty in G. Thus

$$\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger, \qquad \gamma_1 G = c_1^\dagger G, \qquad \|\gamma_1 G\| = 1.$$

The Jordan–Wigner sign is essential: replacing $Z_0 X_1$ by X_1 produces an orthogonal probe because the exact up occupation is $1/2$. This unitary creation preserves the entire initial exterior CAR state: disjoint even observables commute, and exterior odd expectations vanish in both definite-parity states. The origin-down occupation changes from zero to one and total number from four to five. Its mean added energy is $1/2$ with variance $67/256$; this is a localized initial defect with a conserved number flag, not a persistent localization or mass-eigenstate claim. A 27-pulse pointer word prepares $(|0\rangle G + |1\rangle \gamma_1 G)/\sqrt{2}$. Its controlled Majorana is the product of commuting Pauli rotations in $\exp[i\pi(I - Z_{\text{pointer}})(I - \gamma_1)/4]$. This explicitly supplied native qubit service maintains a phase relation between the $N = 4$ and $N = 5$ arms; it does not derive a physical fermion-parity reference from passive repair.

Both arms evolve under the same original Hamiltonian. With a second controlled Majorana and pointer analyzer of phase ϕ,

$$C(t) = \langle \gamma_1 G, e^{-it(H - E_G)} \gamma_1 G \rangle,$$
$$P_0(\phi, t) = \tfrac{1}{2} \left[1 + \mathrm{Re}(e^{i\phi} C(t)) \right].$$

The ground-energy phase is supplied by the evolving reference arm; no separate scalar offset is inserted into the source law. The two executed settings have $t = 1$ and $\phi = 0, \pi/2$. Each setting uses two half-time segments, each with 32 symmetric splitting steps and 337,920 native pulses, followed by a 29-pulse analyzer. Including its ground preparation and pointer, a complete setting uses 681,459 pulses on the full 17-bit field/pointer register.

Protection and complete output accounting. The five C repair seams follow the first time segment, and the five E seams follow the second. The complete repair theorem acts as identity on $N4 \oplus N5$, including cross-number coherences: all native branches vanish there, and the outside branch is identity. This is passive protection during the waits. Preparation and control words are not claimed to remain in this subspace at every intermediate pulse. The final inverse Majorana can create an $N = 6$ component after the charged particle has moved; no repair is applied after that analyzer.

Every source first-export failure is retained as its own terminal record. The numerical execution also retains all 130 repair outcome slots, including off-subspace outputs from control error, and all

six final pointer outcomes. A native repair outcome terminates with that record; continuation uses the outside branch without renormalization. The lossless archive retains the prepared full ground, both full final states and every first-failure weight table. Individual failure vectors remain addressable from their actual source prefixes.

Connection to the resolved spectrum. All field and pointer words are consumed against the exact source templates. Their complete operator errors, the preparation error and the full-Fock Strang errors of both time segments give a complete output error below 3/5000 for either setting. Contractivity through the complete repair and readout channels preserves the telescoping bound. The same bound controls each unconditioned pointer probability. Independently, the 78 certified spectral intervals evaluate

$$C(t) = \sum_{j=1}^{78} w_j e^{-i\Delta E_j t}$$

with rational sine and cosine enclosures. Adding the source error gives certified pointer intervals, checked against the literal execution and an independent exponential of the original Hamiltonian. The $\pi/2$ outcome is separated from $1/2$; the $U = 0$ comparator has the purely real response $\cos(\sqrt{3}t/16)$. Zero-time native runs check the analyzers as well. These two settings measure the charged characteristic function. They do not independently resolve all 78 lines, select an autonomous ground, prepare a thermodynamic vacuum, establish laboratory units, or turn finite energy gaps into relativistic particle masses.

The complete finite matrix spectrum and its exact spatial algebra. The 78-line local down-addition result extends to all eight source columns $V_x = c^\dagger_{x\downarrow} G$. On the entire 560-state invariant sector, exact arithmetic proves $q(16H)V = 0$ for the same monic degree-78 polynomial q and verifies all 78 matrix moments in $\operatorname{span}\{I, A\}$, where $A = A^{\mathsf{T}}$, $A^2 = 3I$ and $\operatorname{tr} A = 0$. Polynomial division of the resolvent then gives

$$\mathcal{G}(z) = \frac{16\big[p_I(x)I + p_A(x)A\big]}{1296q(x)}, \qquad x = 16z - 4\sqrt{3}.$$

The numerator coefficients are exact convolutions of q and those source moments. This is a rational resolvent identity, rather than a truncated large-frequency expansion.

A second exact polynomial calculation gives $q \mid (p_I^2 - 3p_A^2)$. At each simple root x_i, define $\alpha_i = p_I(x_i)/(1296q'(x_i))$ and $\beta_i = p_A(x_i)/(1296q'(x_i))$. The earlier interval certificates give $\alpha_i > 0$; the new identity gives $3\beta_i^2 = \alpha_i^2$, and strict interval signs determine $s_i = \operatorname{sign}\beta_i$. Therefore

$$R_i = \alpha_i I + \beta_i A = 2\alpha_i P_{s_i}, \qquad P_\pm = \tfrac{1}{2}(I \pm A/\sqrt{3}).$$

Both $P_\pm$ have rank four. There are 40 plus-band lines and 38 minus-band lines; each band's $2\alpha_i$ weights sum to one. The rank of $E_i V$ equals that of its Gram matrix R_i, so the reducing cyclic space generated by all eight probes has dimension $78 \cdot 4 = 312$. The remaining 248 source dimensions are orthogonal to these probes and their time translates; their spectrum is not claimed.

Writing $\mathcal{G} = FI + JA$, both band functions have strictly negative imaginary part for $\operatorname{Im} z > 0$. Hence

$$\mathcal{G}^{-1} = \frac{FI - JA}{F^2 - 3J^2}, \qquad \Sigma = zI - (A/16 + I/2) - \mathcal{G}^{-1}.$$

The full finite inverse response and memory are exactly onsite plus the original signed nearest-neighbor adjacency. This special cube algebra does not supply an infinite-volume spatial limit or a Lorentz metric.

The certified common spectral support lies strictly between $g = 299/10000$ and $b = 37/20$. On the reducing cyclic space, both $K = H - E_G$ and its compression perpendicular to the observed columns obey these bounds. The exact coupling $W = KV - VM_1$ lies in that same space, so the orthogonal unobserved sector contributes zero to the Schur memory. Using $W^\dagger W = I/4$, its order-p Taylor remainder satisfies

$$\|R_p\| \le \frac{r^{p+1}}{4g^{p+1}(g-r)}, \qquad |z| \le r < g.$$

This improves the earlier conservative full-sector disk without assigning the stronger gap to the unresolved sector or to infinite volume. Evaluation at $x = -4\sqrt{3}$ now computes $-\mathcal{G}(0)$ and $\Sigma(0)$ exactly over $\mathbb{Q}(\sqrt{3})$, including the previously bounded coefficients 12.9058381883 and -4.9026412320. No floating linear solve, native detector execution, physical species assignment or relativistic rest-mass inference is used in this extension.

Excluding a complete finite-gap mass dictionary. The full support gives an exact test of assigning these visible single-charge gaps to charged-lepton rest energies under one positive unit factor. Its largest-to-smallest ratio is below 62, including the positive up-channel line. The retained retrospective PDG 2025 benchmark has muon/electron ratio above 206 and tau/electron ratio above 3477, even in the declared central-value plus/minus reported-uncertainty boxes. If a source ratio is at most R and the target ratio at least $Q > R$, simultaneous relative errors at most $\epsilon < 1$ require

$$\frac{1-\epsilon}{1+\epsilon} \le \frac{R}{Q}, \qquad \epsilon \ge \frac{Q-R}{Q+R}.$$

The exact rational calculation therefore forces an error above 54 percent for at least one member of an electron/muon assignment, and above 96 percent for an electron/tau assignment, for every common positive scale and every choice of visible lines. The target values do not enter the source law or spectrum. This rejects the stated finite-gap mass dictionary, not a continuum, an unobserved or composite sector, or OPH as a whole. The finite gaps are not thereby proved to be relativistic rest masses, and no eligible prospective comparison is added.

A local inverse response on the interacting infinite ground. The original zero-chemical-potential Weyl/contact action on $\mathbb{Z}^3$ now admits a controlled spatial inverse response on the previously constructed thermodynamic ground, at fixed spacing and away from real frequency. This does not extrapolate the special two-eigenvalue cube identity. Let ω be the translation-averaged full-ground limit, L_ω its positive GNS generator, and

$$L = L_\omega \oplus -\overline{L_\omega}, \qquad V_{x\sigma} = c^\dagger_{x\sigma}\Omega \oplus \overline{c_{x\sigma}\Omega}.$$

The CAR give an isometry V from $\ell^2(\mathbb{Z}^3;\mathbb{C}^2)$. Define J in the same way from $b_{x\sigma} = n_{x\bar\sigma}c_{x\sigma}$ and write $\rho_{x\sigma\tau} = \omega(c^\dagger_{x\tau}c_{x\sigma})$. Exact onsite CAR identities, with vanishing different-site anticommutators, give

$$R = \bigoplus_x (\mathrm{tr}(\rho_x)I - \rho_x), \qquad V^\dagger J = J^\dagger V = J^\dagger J = R,$$
$$LV = Vh + UJ, \qquad 0 \le R \le I, \quad \|LV\| \le \sqrt{3}c + U.$$

Finite-support approximants and closedness of L extend the last identity to the entire one-particle input space. With $P = VV^\dagger$, $Q = I - P$, the bounded off-diagonal coupling is $W = QLV =$

$U(J-VR)$. Subtracting $WV^\dagger + VW^\dagger$ from L proves self-adjointness of its compressed complement $T = QLQ|_{Q\,\mathrm{Dom}\,L}$. The exact infinite-space Schur identity is therefore

$$\begin{aligned} \mathcal{G}(z) &= V^\dagger (z-L)^{-1} V = [zI - h - UR - \Sigma(z)]^{-1}, \\ \Sigma(z) &= W^\dagger (z-T)^{-1} W, \qquad\qquad W^\dagger W = U^2(R-R^2). \end{aligned}$$

In particular $\|\Sigma(z)\| \le U^2/(4\,\mathrm{Im}\,z)$.

This ground has strictly nonzero memory for $U > 0$. The source hidden-spin frame obeys $R_{x+t} = \pm R_t R_x$, so translations normalize its global SU(2) symmetry. Every full-box ground projector and the exterior vacuum are invariant; translated averages and weak-star limits retain the symmetry. Hence $\rho_x = rI$, with the earlier density theorem giving $1/8 \le r \le 1/2$, and

$$\frac{7U^2}{64} I \le W^\dagger W = U^2 r(1-r) I \le \frac{U^2}{4} I.$$

The spectral theorem gives $iy\Sigma(iy) \to W^\dagger W$ strongly as $y \to \infty$. Thus the same-mode response cannot be that of a number-conserving static quadratic Hamiltonian with zero memory. Neither a scattering theorem nor faithfulness of the local ground-state density matrices follows.

A quantitative connected four-point witness. The same ground has a non-Gaussian multi-time field, a stronger statement than nonzero two-point memory. Fix $a = c_{x\sigma}$ and $n = n_{x\bar{\sigma}}$, and set

$$A(t) = \{\tau_t(a), a^\dagger\}, \qquad D(t) = A(t) - \omega(A(t))I, \qquad F(t) = \omega(D(t)^\dagger D(t)).$$

The local commutator gives $D(0) = 0$ and $D'(0) = -iU(n - rI)$. Since $n^2 = n$, $F(t)/t^2 \to U^2 r(1-r) \ge 7U^2/64$. An explicit remainder uses $M = 9c^2 + 12cU + U^2$ and $\delta = [H, \cdot]$:

$$\|\delta^2(a)\| \le 3c(3c+U) + U(6c+3c+U) = M, \qquad \|D(t) + itU(n - rI)\| \le 2Mt^2.$$

Thus, for $U > 0$,

$$F(t) \ge \frac{U^2 t^2}{64} > 0, \qquad 0 < |t| \le \frac{U}{16M}.$$

For odd zero-mean operators define the connected fermionic function

$$\begin{aligned} \kappa_4(B_1, B_2, B_3, B_4) = {} & \omega(B_1B_2B_3B_4) - \omega(B_1B_2)\omega(B_3B_4) \\ & + \omega(B_1B_3)\omega(B_2B_4) - \omega(B_1B_4)\omega(B_2B_3). \end{aligned}$$

With $(B_1, B_2, B_3, B_4) = (a, \tau_t(a)^\dagger, \tau_t(a), a^\dagger)$, $F(t)$ is the sum of κ_4 on words $1234, 1243, 2134, 2143$. All crossed contractions, including anomalous ones, cancel pairwise. At least one connected function therefore has magnitude $U^2t^2/256$ or greater on the displayed interval. This excludes a quasifree representation of the complete multi-time field by linear fermionic fields, including pairing and auxiliary Gaussian modes. Indeed any quadratic evolution on an enlarged CAR algebra preserves linearity of an observed mode, making its anticommutator scalar and $D(t) = 0$ in every reference state. Thus even a non-Gaussian reference cannot realize this witness through such a linear quadratic embedding. It does not exclude nonlinear encodings or establish continuum scattering.

The full ordered Euclidean hierarchy. The same positive ground generator $L_\omega \geq 0$ defines, for ordered real times $s_1 \leq \cdots \leq s_n$ and bounded local CAR operators,

$$S(A_1, s_1; \ldots; A_n, s_n) = \langle \Omega, A_1 e^{-(s_2-s_1)L_\omega} A_2 \cdots e^{-(s_n-s_{n-1})L_\omega} A_n \Omega \rangle, \qquad |S| \leq \prod_j \|A_j\|.$$

Every finite approximant uses one common translated full-ground state and its positive generator $H_\Lambda - E_\Lambda$ for the whole word, with zero exterior Hamiltonian, before averaging over translations. For each positive gap the spectral theorem gives

$$e^{-sL} = \int_{\mathbb{R}} \frac{s}{\pi(s^2+t^2)} e^{itL}\, dt, \qquad L \geq 0.$$

The product integral is dominated by $\prod_j \|A_j\|$. Its real-time integrand is the expectation of $A_1 \tau_{t_1}(A_2) \tau_{t_1+t_2}(A_3) \cdots$. Local-dynamics bounds, telescoping the product, and the same weak-star ground subsequence give uniform convergence on compact time boxes. The omitted product-measure tail is at most $\sum_j 2s_j/(\pi T_j)$, capped by one; a finite-versus-limit comparison costs twice this tail times the product of operator norms. Dominated convergence therefore supplies every ordered Euclidean n-point limit on this one ground, without another subsequence or a state-convergence rate. Zero gaps are handled by the identity operator.

For a positive-time word $F = (A_1, t_1; \ldots; A_n, t_n)$ define

$$\Psi_F = e^{-t_1 L_\omega} A_1 e^{-(t_2-t_1)L_\omega} A_2 \cdots e^{-(t_n-t_{n-1})L_\omega} A_n \Omega.$$

The reflected list is $(A_n^*, -t_n; \ldots; A_1^*, -t_1)$, in that order. Concatenation with a second positive-time word gives $S(\theta F, G) = \langle \Psi_F, \Psi_G \rangle$; hence every reflected correlation matrix is positive semidefinite. No CAR factors are reordered. Exact rational two-mode controls check the Gram identity, all 31 principal minors of a five-word example, and a negative odd-word cross entry. This proves ordered time-reflection positivity at fixed spacing. It does not establish $O(4)$ covariance, bosonic permutation symmetry, a spatial continuum, or the full relativistic Euclidean axioms.

Uniform inverse and spatial decay bounds. Fix $\operatorname{Im} z \geq \eta > 0$, $|z| \leq Z$, with $Z \geq \eta$. For a unit vector $v = Vf$, Cauchy–Schwarz applied to $\langle (\bar{z} - L)v, (z-L)^{-1}v \rangle = 1$ yields

$$-\operatorname{Im} \mathcal{G}(z) \geq B_0^{-1} I, \qquad \|\mathcal{G}(z)^{-1}\| \leq B_0, \qquad B_0 = \frac{(Z + 2c + U)^2}{\eta}.$$

Both $\mathcal{G}$ and its adjoint are bounded below; its range is therefore closed and dense, which proves surjectivity in infinite volume. No determinant or finite-volume gap is used.

Let $d = \|x - y\|_\infty$. The original graded locality bound gives

$$\|\mathcal{G}_{xy}(z)\| \leq A(1+d)^2 e^{-2\mu d},$$
$$\mu = \min\{1/8, \eta/(384c)\}, \qquad A = 192c/\eta^2 + 8/\eta.$$

For $d \geq 2$, approximate the evolved odd operator in the radius $d-1$ cube, whose anticommutator with the remote mode is zero. Split the Laplace integral at $(d-1)/(192c)$ and use the previous full-CAR boundary estimate on the first part and the uniform norm bound on its tail. This proves the displayed constants without assuming spatial clustering of ω.

Put $q = (1+\mu)^{-1}$ and $T_n(q) = \sum_{d \geq 1} d^n q^d$. These are rational functions with positive Eulerian coefficients. Define

$$\widehat{S} = 24T_5(q) + 48T_4(q) + 26T_3(q) + 4T_2(q) + 2T_1(q),$$
$$\alpha = \min\{\mu, (2B_0 A \widehat{S})^{-1}\} > 0.$$

For any bounded real one-Lipschitz function f, the block Schur test gives

$$\|e^{\alpha f}\mathcal{G}e^{-\alpha f}-\mathcal{G}\|\le\alpha A\widehat{S}\le(2B_0)^{-1}.$$

A Neumann series bounds the conjugated inverse by $2B_0$. Choosing $f(u)=\min\{\|u-y\|_\infty,R\}$ with $R\ge d$ proves

$$\begin{aligned}\|(\mathcal{G}^{-1})_{xy}(z)\|&\le 2B_0e^{-\alpha d},\\ \|\Sigma_{xy}(z)\|&\le(Z+U+2B_0+2c)e^{-\alpha d}.\end{aligned}$$

Only bounded weights were used. The constants are conservative bounds, not estimates of a physical correlation length.

The controlled spatial symbol and its geometric boundary. Translation invariance and exponential summability give an analytic two-by-two inverse-response symbol $K(k,z)$ for $\|\operatorname{Im}k\|_1<\alpha$. For real momentum increment p, its total-degree-n Taylor remainder is bounded by

$$\frac{2B_0\|p\|_1^{n+1}}{(n+1)!}\sum_{x\in\mathbb{Z}^3}\|x\|_\infty^{n+1}e^{-\alpha\|x\|_\infty}.$$

The shell count $24d^2+2$ and the same rational T_n functions give explicit upper bounds on this convergent sum. A frequency circle of radius $\eta/2$ gives, for $|w|<\eta/2$, the additional remainder $B_*(2|w|/\eta)^{n+1}/(1-2|w|/\eta)$, where B_* is the above inverse bound evaluated at $(\eta/2,Z+\eta/2)$.

These estimates supply actual local spatial coefficients on the interacting infinite ground, with a controlled complex-frequency domain. They do not compute the ground functional or those coefficients, and deteriorate as $\eta\to0$. A real hyperbolic principal symbol, common matter coefficients and full action normalization are still required before applying the existing `wave_density_metric` and `source_tetrad` interfaces. The joint modular, area, stress and Einstein balance requirements remain separate.

For a refinement $c=v/a$ at fixed $v,U,Z,\eta>0$, the retained estimates have $\alpha=O(a^9)$ and therefore physical momentum width $\alpha/a=O(a^8)$; the connected-witness time window is $O(a^2)$. These shrinking bounds do not prove uniform continuum control or a no-go theorem for the action.

The full ground-family spectrum and its internal spin. The complete five-dimensional $N=4$ ground space closes a remaining finite spectral ambiguity. Work in the exact hidden-spin frame, with G_0 its spin-two highest-weight ground, $\lambda=\sqrt{3}/16$, and $V_{\uparrow,x}=c_{x\uparrow}^\dagger G_0$, $V_{\downarrow,x}=c_{x\downarrow}^\dagger G_0$. The actual 56×560 raising matrix $T=S_+$ obeys

$$TV_\downarrow=V_\uparrow,\qquad TT^\dagger=5I,\qquad H_{5,0}T=TH_{4,1}.$$

The full spatial Gram is $V_\uparrow^\dagger V_\uparrow=P_+$, and $(H-E_G)V_\uparrow=\lambda V_\uparrow$. At magnetic weight $3/2$,

$$V_{\text{high}}=T^\dagger V_\uparrow/5,\qquad V_{\text{low}}=V_\downarrow-V_{\text{high}},\qquad TV_{\text{low}}=0,\qquad V_{\text{high}}^\dagger V_{\text{high}}=P_+/5.$$

The retained residue at λ is exactly $P_+/5$. Positivity therefore leaves no low-spin contribution at that line: it has total hidden spin $5/2$, while the other 77 visible lines have total hidden spin $3/2$.

This conclusion covers every ground vector. Put $G_k=S_-^kG_0$, with $\|G_k\|^2=k!4!/(4-k)!$, $0\le k\le4$. Exact occupation-space CAR give, for every site,

$$\begin{aligned}c_\downarrow^\dagger G_k&=S_-^{k+1}V_\uparrow/5+S_-^kV_{\text{low}},\\ c_\uparrow^\dagger G_k&=(5-k)S_-^kV_\uparrow/5-kS_-^{k-1}V_{\text{low}}.\end{aligned}$$

The second term vanishes at $k = 0$. All eighty creator columns are checked on the original source; the Hamiltonian commutes with every ladder. No coherent ground superposition or linear CAR probe can introduce another addition frequency.

The full $N = 5$ space has spin multiplicities $56, 504, 1008$ at $S = 5/2, 3/2, 1/2$, respectively. Four reachable copies occur at λ, and four at each of the other 77 lines. Thus the complete reachable dimension is

$$4 \cdot 6 + 77 \cdot 4 \cdot 4 = 1256 \quad \text{inside} \quad \binom{16}{5} = 4368.$$

The dark complement has dimensions $312, 784, 2016$ in those three spin sectors. These are dimensions of source subspaces, not full energy degeneracies; their dark eigenvalues remain uncomputed. All annihilator probes similarly generate sixteen $N = 3$, spin-3/2 states with positive removal cost λ, leaving 544 dark dimensions.

The invariant mixed-reference energy readout. For $\rho_G = P_G/5$, the exact addition and retarded matrices are

$$\overline{\mathcal{G}}_{\mathrm{add}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda}\right] \otimes I_2,$$
$$\overline{\mathcal{G}}_{\mathrm{ret}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda} + \frac{P_-}{z+\lambda}\right] \otimes I_2.$$

At each site and fixed spin, the addition line λ has weight $3/10$, and each other line has half its previous weight. The total addition weight is $3/4$; conditional weights are therefore $2/5$ at λ and $2w_i/3$ elsewhere. Removal has weight $1/4$ at the signed retarded pole $-\lambda$. All 78 positive lines survive. The 79 signed poles have rank-eight residues, giving a minimal mixed-state response realization of dimension 632. This realization keeps sixteen observed columns and differs from the physical reachable range that keeps all five ground inputs distinct.

Direct full-source Gram matrices verify every spin and spatial entry of the zeroth and first moments, including

$$\overline{M}_{0,\mathrm{ret}} = I_{16}, \qquad \overline{M}_{1,\mathrm{ret}} = (A \otimes I_2)/16 + I_{16}/4.$$

The actual physical-frame block is $g_{xy} R_x^\dagger R_y$; intersite spin coherence is retained, while local blocks are scalar. The same positive-energy envelope and finite charged-lepton ratio exclusion now hold for every ground choice and linear single-CAR probe. These total internal spins include the spin-two background. They do not identify intrinsic particle or Lorentz spin, a measured species, or a relativistic mass. The result does not exclude composite probes or a continuum spectrum, and does not execute native preparation or complete finite-shot spectroscopy of the mixed reference.

Complete finite-resolution energy readout. For a unit one-particle probe f, the same ground defines the probability measure on signed addition and removal energies

$$\mu_f(B) = \langle Vf, 1_B(L) Vf \rangle.$$

The complete Kraus pair $c(f)^\dagger, c(f)$ labels addition and removal; each branch has a nonnegative physical excitation energy. Only the response-frequency variable assigns a minus sign to removal. The exact identities $LV = Vh + UJ$ and $V^*J = J^*J = rI$ give the operator moments

$$M_0 = I, \qquad M_1 = h + UrI, \qquad M_2 = h^2 + 2Urh + U^2 rI.$$

The second-moment identity is a quadratic-form statement and does not require $Vf \in \operatorname{Dom} L^2$. For a local mode at $c = 1/8, U = 1$, the mean lies in $[1/8, 1/2]$, the second moment in $[19/128, 67/128]$, and the variance in $[17/128, 35/128]$. Thus $\mu_f(|E| \geq R) \leq 67/(128R^2)$. More generally, $\|hf\| \leq K$ gives a second moment at most $K^2 + UK + U^2/2$, useful for uniformly controlled physical smearing.

Choose $M \geq 2$ and $\tau > 0$. The positive effects

$$K_k(E) = \left| \frac{1}{M} \sum_{j=0}^{M-1} e^{-ij(\tau E - 2\pi k/M)} \right|^2, \qquad q_k = \int K_k(E)\, d\mu_f(E), \quad 0 \leq k < M,$$

sum exactly to one by finite Fourier orthogonality. Writing $\phi_f(t) = \int e^{-itE}\, d\mu_f(E)$, their complete probabilities are

$$q_k = \frac{1}{M} + \frac{2}{M} \sum_{n=1}^{M-1} \left(1 - \frac{n}{M}\right) \operatorname{Re}(e^{2\pi ikn/M} \phi_f(n\tau)).$$

The principal energy centers are modulo $2\pi/\tau$ and are readout bins, not asserted eigenvalues. For $0 < \tau\Delta \leq 3$, circular error has probability at most $\min\{1, 10/(M\tau^2\Delta^2)\}$, improved to $22/(7M\tau\Delta - 44)$ when its denominator is positive. This follows by summing the two inverse-square grid tails. A second-moment bound C adds an alias term at most $C/(3/\tau - \Delta)^2$. Consequently the probability of a real energy error at least Δ is controlled, without a pointwise spectral density or an isolated-pole assumption.

The complete frozen 78-line cube measure now gives certified distributions for $M = 8, 16, 64$ at $\tau = 1$. At $\Delta = 1/2$, direct positive-kernel and Fourier interval calculations bound the respective error probabilities by 0.100284517428, 0.055193158232, and 0.020209031434. All outcomes are retained. The previously executed native time-one quadratures check the first harmonic; higher harmonics here are source calculations, not new native long-time or finite-shot measurements.

Along the existing full-ground subsequence, the finite positive spectral measures converge weakly. The uniform fixed-spacing spatial kernel bound also promotes entrywise convergence to strong convergence of the compressed resolvents on the common one-particle space. No identification of the varying GNS Hilbert spaces is assumed. An explicit finite readout error adds the local-dynamics, native-control, and statistical errors to the still-unspecified finite-ground state discrepancy; the last term is never set to zero. Sharp interval probabilities require zero limiting mass at their endpoints. These results do not supply physical particle identifications, relativistic rest masses, or an interacting continuum.

An all-valley positive-energy continuum field net. There is now a spacing-limit construction beyond fixed-lattice response. Take $c(a) = v/a$, fixed $v > 0$, and $0 \leq U(a) \leq U_{\max} < \infty$. At each spacing retain the full original Weyl/contact dynamics and a translation-invariant full ground. On centered cells $C_{a,n} = a(n + [-1/2, 1/2)^3)$ define

$$(J_a f)(n) = a^{-3/2} \int_{C_{a,n}} f(x)\, dx, \qquad P_a = J_a^* J_a.$$

Cellwise Cauchy–Schwarz and coordinate averaging give $\|J_a\| \leq 1$ and $\|P_a f - f\| \leq a \sum_j \|\partial_j f\|$ for $f \in H^1$. Thus $P_a \to I$ strongly. For every $b \in \{0,1\}^3$ set $M_b u(n) = (-1)^{b\cdot n} u(n)$ and $C_{a,b}(f) = c(M_b J_a f)$, with the conventional antilinear annihilator. All eight valleys evolve under the original full Hamiltonian. Writing $D_{a,j} f = [f(\cdot + ae_j) - f(\cdot - ae_j)]/(2a)$ gives

$$h_a M_b J_a f = M_b J_a \left[iv \sum_j (-1)^{b_j} \sigma_j D_{a,j} f \right], \qquad \|h_a M_b J_a f\| \leq v \sum_j \|\partial_j f\|.$$

For $b \neq d$, shifting one lattice step in a coordinate where they differ reverses the cross-valley pairing. Subtracting the shifted pairing proves

$$|\langle M_b J_a f, M_d J_a g\rangle| \leq \frac{a}{2}\left(\|\partial_j f\|\|g\| + \|f\|\|\partial_j g\|\right).$$

The limiting equal-time fields therefore have the full continuum CAR over $L^2(\mathbb{R}^3;\mathbb{C}^2)\otimes\mathbb{C}^8$.

The necessary time estimate holds in the full CAR operator norm. For $B(u)=\sum_{x\sigma}\overline{u_{x\sigma}}n_{x\bar{\sigma}}c_{x\sigma}$, the onsite anticommutator is positive and obeys $\{B(u_x),B(u_x)^*\}\leq\|u_x\|^2 I$. Different-site odd anticommutators vanish, hence $\|B(u)\|\leq\|u\|$. The exact contact commutator consequently gives

$$\|\tau_{a,t}(C_{a,b}(f))-\tau_{a,s}(C_{a,b}(f))\|\leq|t-s|\left(v\sum_j\|\partial_j f\|+U_{\max}\|f\|\right).$$

This avoids an extensive interaction-norm estimate.

A diagonal subsequence of spacings makes every word in a countable dense set of smooth smears, rational times and all eight valleys converge simultaneously. The displayed time and norm estimates extend the limits to all L^2 smears and real times. Polynomial positivity, norm bounds and the CAR pass to the limit and construct bounded fields in one GNS representation. Common time shifts are strongly continuous. Nearest *even* lattice translations approximate every physical translation while keeping all valley phases equal to one, so continuous spatial translations also act on the limiting net. For every multi-time polynomial X, the correlations $\omega_a(X_a^*\tau_{a,t}(X_a))$ have nonnegative energy support and converge uniformly on compact time intervals. Testing against smooth negative-energy spectral functions and using dominated convergence therefore proves a nonnegative generator on the entire limiting GNS space, not only on linear-field vectors.

Graded locality also survives with an explicit bound. Suppose f is supported in $[-r,r]^3$, g outside $[-R,R]^3$, and put $\delta=(R-r)/2-48vT>0$. For $0<a\leq\min\{1,(R-r)/8\}$ and $|t|\leq T$, the original full-CAR bound, with lattice radii $\lceil r/a\rceil+1$ and $\lfloor R/a\rfloor-1$, yields

$$\begin{aligned}\|\{\tau_{a,t}(C_{a,b}(f)),C_{a,d}(g)^{\#}\}\|&\leq 192vT(2r+5)^3(2R+1)^2\|f\|\|g\|a^{-6}e^{-\delta/a}\\&\leq 192\cdot 7!vT(2r+5)^3(2R+1)^2\|f\|\|g\|\frac{a}{\delta^7}.\end{aligned}$$

Here $\#$ allows an adjoint or no adjoint. Inserting this norm estimate between arbitrary polynomial vectors makes the limiting anticommutator the zero operator. A finite partition of unity extends the result to separated compact supports. The continuum net thus has exact graded locality at sup-norm distance greater than $96v|t-s|$. This is a conservative propagation cone, not a Lorentz metric.

There is a further uniform response bound on every physical momentum window $|p|\leq P$ around any retained valley. These windows reduce the translation-invariant response and have kinetic norm at most vP. The exact Schur formula gives

$$\|\mathcal{G}_a(z)^{-1}\|_P\leq|z|+vP+\frac{U_{\max}}{2}+\frac{U_{\max}^2}{4\operatorname{Im}z},\qquad -\operatorname{Im}\mathcal{G}_a(z)\geq\frac{\operatorname{Im}z}{(|z|+vP+U_{\max})^2}I_P.$$

These bounds are uniform in spacing away from real frequency. The cell-average map differs from the exact band isometry by at most $a^2P^2/24$, so the two conventions have the same limiting correlations. Weak convergence alone does not allow interchanging the inverse and limit.

A norm-controlled free limit and the remaining interacting step. A Duhamel comparison with the free lattice evolution uses only $\|B(u)\| \le \|u\|$ and gives an error at most $|U(a)t|\|u\|$. The central-difference remainder then gives, for $f \in H^3$ and $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$,

$$\|\tau_{a,t}(C_{a,b}(f)) - C_{a,b}(e^{ith_b}f)\| \le |t|\left(U(a)\|f\| + \frac{va^2}{6}\sum_j \|\partial_j^3 f\|\right).$$

If $U(a) \to 0$, the limiting dynamics is exactly the free eight-valley Weyl dynamics. Positive energy forces positive-band annihilators and negative-band creators to annihilate the ground. The one-particle zero-energy kernel is zero in L^2; spectral-band approximation and the CAR recurrence therefore uniquely determine the filled-sea quasifree state and all its moments. This is a controlled free continuum, not the required interacting one.

For bounded positive limiting U, the construction still gives the full positive-energy continuum field net, but it has not proved that a connected interaction survives, selected a unique limiting state, or derived Lorentz covariance. Its propagation cone and positive energy supply no common physical stress, geometric modular flow, renormalized area stationarity or completed G1–G6 Einstein package.

Ground occupation and energy constraints under refinement. The original zero-chemical-potential ground satisfies a useful uniform constraint before any continuum particle identification. With normalized Brillouin measure, let $\rho(k)$ be its one-body covariance and $P_\pm(k)$ the spectral projectors of the original Weyl hopping. The free filled sea has onsite covariance $I/2$ and double occupation $1/4$. Its finite-box restrictions give the variational bound

$$0 \le e_{\text{kin}} - e_{\text{free}} + Ud = e_U - e_{\text{free}} \le U/4, \qquad d = \omega(n_\uparrow n_\downarrow).$$

Here $e_{\text{free}} = -c\int \sqrt{\sum_j \sin^2 k_j}$; no free reference is substituted for the interacting state. Set

$$w(k) = \operatorname{tr}(P_+\rho) + \operatorname{tr}(P_-(I-\rho)), \qquad D = \int w(k).$$

Then $0 \le w \le 2$ and the kinetic excess is $c\int \sqrt{\sum_j \sin^2 k_j}\, w(k)$. The region with $\sqrt{\sum_j \sin^2 k_j} \le \delta$ has normalized volume at most δ^3 for $0 < \delta \le 1$, since $2\arcsin\delta/\pi \le \delta$. Splitting this integral proves

$$D \le \min\{2, 2\delta^3 + U/(4c\delta)\}, \qquad |n-1| \le D.$$

For $U \le 24c$, choosing $\delta = (U/(24c))^{1/4}$ gives $D \le 8(U/(24c))^{3/4}$. Thus the density tends to one per site, and the per-spin density r tends to $1/2$, when $U/c \to 0$. At $v = 1/8$, $U = 1$, $a = 1/65536$, $c = v/a$ and $\delta = 1/16$, the exact bound is $1023/1024 \le n \le 1$.

The same defect controls every fixed microscopic local word. The free-sea annihilation parts $c(P_+f)$ and $c^\dagger(P_-f)$ of a unit site-basis field have interacting GNS norm at most $\sqrt{D}$. Moving the first annihilation part to the right by CAR gives the free Wick recursion, with two boundary errors at most $\sqrt{D}$. Hence for an even word of length m,

$$|\omega(A_1\cdots A_m) - \omega_{\text{free}}(A_1\cdots A_m)| \le C_m\sqrt{D}, \qquad C_0 = 0, \quad C_m = 2 + (m-1)C_{m-2}.$$

In particular $C_4 = 8$ and $d \to 1/4$. This controls fixed microscopic local observables; shrinking momentum pockets can still carry a full normalized continuum packet, so no Gaussian continuum conclusion follows.

For cell-averaged physical H^1 packets and $c = v/a$, bounded $U(a) \to U_0$ fixes the first spectral moment to $\langle f, h_{\mathrm{W}} f\rangle + (U_0/2)\|f\|^2$ in every weak spectral limit. The uniform second-moment bound supplies the required first-moment tail control. The scalar term is retained without a chemical-potential shift. Second moments need not pass: the positive measures $(1 - n^{-2})\delta_0 + n^{-2}\delta_n$ converge weakly to δ_0 while their variances tend to one. Thus even a nonzero memory Gram need not survive the limiting response. These constraints supply ground and energy inputs, not relativistic rest masses or an Einstein balance. The proof and exact controls are in `model/continuum_ground_constraints`.

A complete native finite-clock energy measurement. The original zero-chemical-potential cube now has a finite native preparation of its invariant ground reference $P_G/5$. Five azimuths $2\pi j/5$ and three polar cosines $-\sqrt{3/5}, 0, \sqrt{3/5}$, with polar weights $5/18, 4/9, 5/18$, twirl the spin-two highest-weight ground exactly. The azimuth sum cancels every nonzero magnetic difference through four, and the three-node quadrature integrates the remaining degree-four binomial weights to $1/5$. A native four-bit selector has weights $[5, 8, 5]$ repeated five times, followed by zero, divided by 90. Its fifteen conditional full-Fock spin rotations and the original inverse frame are executed; all selector and export outcomes are kept. The complete prepared-state error is below 10^{-9}.

At the origin, a native occupation-copy followed by $\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger$ gives the complete charge instrument with Kraus maps $c_1^\dagger, c_1$. Both branches are retained. Two clock bits control original-action evolutions for model times one and two, with the independently known phase from $E_G = -\sqrt{3}/4$. The positive Fourier transform produces

$$K_k(E) = \left| \frac{1}{4} \sum_{n=0}^{3} e^{-in(E - 2\pi k/4)} \right|^2, \qquad \sum_{k=0}^{3} K_k(E) = 1.$$

This is a complete positive energy measurement. Both physical charge branches have nonnegative excitation cost; principal bin centers are defined modulo 2π, and the removal tag retains the sign needed for the retarded frequency. For the invariant ground, the local addition measure is $\mu_\downarrow/2 + \delta_\lambda/4$ and the removal measure is $\delta_\lambda/4$, where $\lambda = \sqrt{3}/16$. The complete frozen 78-line measure therefore determines every joint charge/bin probability without fitting.

All eighteen-bit clock rows and thirty classical selector/charge columns are propagated through the actual native local instruments. The exact column decomposition used for computation discards no quantum coherence or outcome. Controlled hopping and contact words use at most three local work bits; no abstract controlled Hamiltonian is supplied as a primitive. With sixteen and thirty-two Strang steps, the full-Fock split errors sum to $1887/65536$. Including every preparation, control, scalar-phase and Fourier error gives complete state and total-variation error below $3/100$. A longest successful history contains 612666 native pulses. All 3240 joint selector/tag/clock slots, the selector stops and every first-export failure are retained. The independent original-Hamiltonian exponential is a comparator only.

This construction executes coherent histories through model time three and closes the stated finite native preparation-and-energy-POVM task. Four broad bins do not separately resolve seventy-eight lines. It neither identifies physical species nor converts a finite excitation energy into a relativistic rest mass. It also does not prepare the separate centered-action reference, establish infinite-ground state convergence, or supply the remaining geometric premises.

A CPT constraint on the unchanged charged fields and clock. For a real, even, unit scalar Schwartz packet ϕ, average the signed addition/removal measure over all sixteen continuum field components. Writing this probability measure as ν_ϕ, spin trace cancels the Weyl first moment and

its kinetic cross term. The exact finite-spacing sum rules, $r_a \to 1/2$, and weak lower semicontinuity give

$$\int E\, d\nu_\phi(E) = \mu = U_0/2, \qquad \int E^2\, d\nu_\phi(E) \le C_\phi = v^2\|\nabla\phi\|^2 + U_0^2/2.$$

Only the second-moment inequality passes to the limit. Negative response frequency denotes removal; the two charge-tagged excitation energies are nonnegative with respect to the original positive GNS generator L.

Suppose a proposed physical identification supplies an antiunitary Θ preserving the same ground and L, and mapping the sixteen smeared annihilators A_s into their adjoints by a unitary matrix Q:

$$\Theta\Omega = \Omega, \qquad \Theta L\Theta^{-1} = L, \qquad \Theta A_s\Theta^{-1} = \sum_t Q_{st}A_t^*.$$

The real even packet is preserved by the spatial inversion and conjugation in the usual CPT action. For every spectral set B, antiunitarity and unitarity of Q give

$$\sum_s \|P_L(B)A_s^*\Omega\|^2 = \sum_s \|P_L(B)A_s\Omega\|^2.$$

Thus the averaged addition and removal measures agree, ν_ϕ is reflection symmetric, and its first moment is zero. This precise same-field, same-ground, same-clock CPT identification is therefore impossible for $U_0 > 0$. No CPT theorem is applied without its hypotheses: this does not exclude Lorentz covariance alone or other field dictionaries.

With ν_ϕ^- the reflected measure and total variation defined as $\sup_B |\nu_\phi(B) - \nu_\phi^-(B)|$, Cauchy–Schwarz proves

$$\mathrm{TV}(\nu_\phi, \nu_\phi^-) \ge \frac{\mu^2}{C_\phi}, \qquad W_1(\nu_\phi, \nu_\phi^-) \ge 2\mu.$$

There is a bounded odd witness: for $R = C_\phi/(2\mu)$, clipping E to $[-R, R]$ gives f_R with $|E - f_R(E)| \le E^2/(4R)$ and $\int f_R\, d\nu_\phi \ge \mu/2$. Gaussian packets of width ℓ have $C_\phi = 3v^2/(2\ell^2) + U_0^2/2$. At $v = 1/8$, $U_0 = 1$ and $\ell = 1$, the total-variation lower bound is $32/67$; broad packets make it tend to $1/2$. These are rigorous bounds, not computed infinite-ground spectra or executed native measurements.

A scalar charged-field rephasing changes the generator to $L - \kappa Q_{\mathrm{charge}}$ and the signed first moment to $\mu - \kappa$. Consequently $\kappa = U_0/2$ is the unique necessary centering for this CPT condition. It is a change of charged dynamics, not an additive energy-origin change or a rest mass. Positivity on the old ground is not automatic: the empty two-mode ground of $H = n_0 + 3n_1$ has a centered response of zero mean at $\kappa = 2$, while $H - 2N$ has eigenvalues $0, -1, 1, 0$. The new reference and its ground property require proof.

The free $U(a) \to 0$ control has physical addition and removal energies $v|p|$, zero rest masses, and four Weyl chiralities of each sign. A massive Dirac control, $H_D = Z \otimes (\sigma \cdot p) + mX \otimes I$, has $H_D^2 = (|p|^2 + m^2)I$ and zero trace: its mass couples chiralities and does not create the scalar signed-energy shift. Exact full-CAR counterexamples and the conditional proof are retained in `model/continuum_cpt_constraint`; no physical species, interacting Lorentz completion or Einstein premise is thereby supplied.

A resolved canonical contact-force channel. The bounded-contact refinement also resolves a nonlinear operator in the original action. At each site set

$$b_\sigma = n_{\bar\sigma}c_\sigma, \qquad d_\sigma = (2n_{\bar\sigma} - 1)c_\sigma = 2b_\sigma - c_\sigma.$$

The onsite unitary $Z = I - 2n_\uparrow n_\downarrow$ obeys $Zc_\sigma Z = -d_\sigma$, so d is itself an exact CAR family, with $\|d(u)\| = \|u\|$. It is a composite of the original fields, not an added microscopic mode. The unchanged Hamiltonian satisfies

$$[H, c(u)] = -c((h + U/2)u) - (U/2)d(u).$$

Thus retaining the scalar term does not justify dropping the composite. For $B(u) = \sum_{x,\sigma} \overline{u_{x\sigma}} b_{x\sigma}$, exact local algebra gives

$$\{B(u), c(v)^*\} = \{B(u), B(v)^*\} = Q(u, v) = \sum_x \mathrm{d}\Gamma(q_x), \qquad q_x = (u_x^* v_x)I - v_x u_x^*.$$

Each q_x has rank at most one and $\|q_x\| = \|q_x\|_{\mathrm{HS}} = \|u_x\|\|v_x\|$. The anomalous mixed anticommutator is the pair sum

$$P(u, v) = \{B(u), c(v)\} = \sum_x \overline{u_{x\downarrow} v_{x\uparrow} - u_{x\uparrow} v_{x\downarrow}} c_{x\uparrow} c_{x\downarrow}.$$

Let D_a be the occupation defect above, $L = \|u\|\|v\|$, and $I_4 = \sum_x \|u_x\|^2 \|v_x\|^2$. The uniform four-field Wick comparison, $\omega_a(Q) = r_a\langle u, v\rangle$, and the free covariance identity $\mathrm{Var}_{\mathrm{free}}(\mathrm{d}\Gamma(q)) = \mathrm{Tr}[\Pi q^*(I - \Pi)q] \le I_4$ imply

$$\omega_a(Y_a^* Y_a) \le 4\big[I_4 + (8\sqrt{D_a} + D_a/2)L^2\big]$$

for either $Y_a = 2Q - \langle u, v\rangle I$ or $Y_a = 2P$. For $u = M_b J_a f$, $v = M_e J_a g$ with bounded smooth compactly supported smears, $I_4 \le a^3 \|f\|_\infty^2 \|g\|^2$ and the coefficient supremum is at most $a^3 \|f\|_\infty \|g\|_\infty$. These ground fluctuations therefore vanish.

The passage is an operator statement on the full field hierarchy. A degree-m CAR polynomial obeys $\|[\mathrm{d}\Gamma(q), A]\| \le m\|q\|\|A\|$ for $q = q^*$: its one-body unitary orbit has frequencies in $[-m\|q\|, m\|q\|]$, and the Bernstein interpolation bound applies. The same argument gives $\|[\sum_x n_{x\uparrow} n_{x\downarrow}, A]\| \le m\|A\|$, with degree at most $m + 2$ after the commutator. In the free interaction picture the order-N Dyson remainder is consequently bounded by

$$\|A\|(2U_{\max}|s|)^N \frac{(m/2)_N}{N!}.$$

This is uniform in spacing and volume. Divide $|t| \le T$ into $K = \max(1, \lceil 8U_{\max} T\rceil)$ intervals and choose $N = m + \ell$ at each step. The resulting approximation has degree at most

$$M = 3^K(m_{\mathrm{initial}} + \ell - 1) - (\ell - 1)$$

and relative norm error at most $(1 + 2^{-\ell})^K - 1$. The quadratic coefficient bound then proves $\|[Y_a, X_a]\| \to 0$ for every finite-time field polynomial X_a; choosing $2^{-\ell} \le a^3/(2K)$ gives an explicit $O(a^3 \log(1/a))$ bound at fixed time. Therefore

$$\|Y_a X_a \Omega_a\| \le \|[Y_a, X_a]\| + \|X_a\|\,\|Y_a \Omega_a\| \longrightarrow 0.$$

The same argument covers initial composite fields, whose degree is three.

Adjoining their equal-time smearings to the common subsequence constructs operators $D_b(f)$ alongside the original $C_b(f)$, with

$$\{D_b(f), D_e(g)^*\} = \delta_{be}\langle f, g\rangle I, \qquad \{D_b(f), C_e(g)^*\} = \{D_b(f), C_e(g)\} = 0.$$

Together with the original CAR and vanishing same-family anomalous anticommutators, these are two copies of the eight-valley CAR algebra. This is algebraic independence; the state is not asserted to factorize or obey Wick's rule. The original correlations and their cyclic positive-energy representation are preserved. Its continuous time action is not automatically extended to the entire enlarged GNS space: the composite derivative contains kinetic terms of order $1/a$. Thus the result resolves the contact-force channel but does not determine whether its bounded-frequency weight survives, escapes to infinite energy, or produces an interacting continuum. It supplies no physical species, Lorentz stress or geometric modular identification. A positive quadratic two-mode control makes this boundary explicit: for $h = \left(\begin{smallmatrix} E_0 & \kappa \\ \kappa & \Lambda \end{smallmatrix}\right)$, with $E_0\Lambda > \kappa^2$ and $\Lambda > E_0 > 0$, the memory Gram stays κ^2 but $\Sigma(z) = \kappa^2/(z-\Lambda) \to 0$. The field error from free energy E_0 is at most $(2\kappa+\kappa^2|t|)/(\Lambda - E_0)$. The full proof and exact controls are in `model/continuum_composite_channel`.

Regional modular data on the full particle ground. The invariant finite reference $\rho_G = P_G/5$ now supplies regional modular data on the same ground used for the particle spectrum. In the hidden spin frame, its highest-weight vector is the Slater determinant of four negative spatial orbitals, with projector $P_- = (I - A/\sqrt{3})/2$. Averaging a common spin rotation yields ρ_G. For a region R, let λ_j be the eigenvalues of the spatial compression $(P_-)_R$. The Slater Schmidt decomposition and the spin average give, for every occupied natural-orbital subset I,

$$w_I = \prod_{j\in I} \lambda_j \prod_{j\notin I} (1-\lambda_j), \qquad \rho_{R,I} = \frac{w_I}{|I|+1} P_{\mathrm{sym},I}. \tag{1}$$

Here $P_{\mathrm{sym},I}$ projects onto the symmetric spin space of dimension $|I|+1$; all other states have zero weight. Indeed phase integration deletes unequal spin counts, and the remaining diagonal integral is $\binom{k}{l}\int_0^1 t^{k-l}(1-t)^l\,dt = 1/(k+1)$. Thus, retaining the complete kernel,

$$S(\rho_R) = \sum_j h_2(\lambda_j) + \sum_I w_I \log(|I|+1). \tag{2}$$

For $0 < \lambda_j < 1$ the modular generator on the positive support is

$$K_R = -\sum_j \log(1-\lambda_j) I + \sum_j \log\frac{1-\lambda_j}{\lambda_j} N_j + \log(N_R+1). \tag{3}$$

Endpoint occupations are handled as fixed blocks, without taking logarithms of zero. This is a support identity, not a full-CAR faithfulness claim.

Every adjacent two-site region has positive spectrum

$$\left\{ \frac{1}{6}\ [1],\quad \frac{2-\sqrt{3}}{12}\ [2],\quad \frac{2+\sqrt{3}}{12}\ [2],\quad \frac{1}{18}\ [3] \right\}, \tag{4}$$

where brackets denote multiplicities. Its eight other eigenvalues are zero. The support consists of the vacuum, the four one-particle states, and the three two-particle triplet states. With H_R the actual internal regional Hamiltonian, whose contact term vanishes on this support,

$$K_R = \log 6\, I + \beta_R H_R + \log(N_R+1), \qquad \beta_R = 16\log(2+\sqrt{3}). \tag{5}$$

Exact signed CAR traces of all five source ground vectors verify this identity on all twelve cube bonds; the sixteen nonadjacent pairs give $K_R = \log 4\, I + \log(N_R+1)$ instead. The same local

spin-frame change conjugates the density, support and Hamiltonian back to the original source coordinates.

For every Hermitian trace-zero variation δ within the positive support, including noncommuting variations,

$$DS_{\rho_R}(\delta) = \beta_R \operatorname{Tr}(\delta H_R) + \operatorname{Tr}\big(\delta \log(N_R + 1)\big). \tag{6}$$

The second term is central on the supported number-preserving algebra, not on the full CAR algebra. Since $\lambda^+_{\min} = (2-\sqrt{3})/12 > 1/48$, for $\delta = \sigma - \rho_R$ with σ supported there and $0 \le \epsilon < 1$,

$$0 \le D(\rho_R + \epsilon\delta \| \rho_R) \le \frac{24\epsilon^2 \|\delta\|^2_{\mathrm{HS}}}{1-\epsilon}. \tag{7}$$

In contrast, a density $\sigma_\perp$ in the orthogonal kernel obeys the full-state identity $S((1-\epsilon)\rho_R + \epsilon\sigma_\perp) = (1-\epsilon)S(\rho_R) + \epsilon S(\sigma_\perp) + h_2(\epsilon)$, whose derivative diverges at zero. The kernel is therefore retained in the operational domain; no supported first law is applied to such a disturbance. These are same-ground finite modular and entropy inputs. Identifying their central term with a geometric edge, their energy pairing with a diamond charge, or their coefficient with a physical clock still requires the existing Einstein theorem's common geometric and scaling premises.

Deriving a centered clock and constructing its new ground. The CPT obstruction motivates a separate, explicitly changed charge clock. On a bipartite Weyl lattice with V sites, the one-particle antiunitary $J = \eta(i\sigma_y)\mathcal{K}$, where $\eta_x = (-1)^{x_1+x_2+x_3}$ and $\mathcal{K}$ is conjugation, anticommutes with the hopping. Its canonical particle-hole transform satisfies

$$\mathcal{P}(N) = 2V - N, \qquad \mathcal{P}(H) = H + U(V-N), \qquad \mathcal{P}(H-\kappa N) - (H-\kappa N) = (U-2\kappa)(V-N).$$

Thus $K = H - UN/2$ is the unique scalar charge centering invariant on the full Fock algebra. This changes charged dynamics; the original Hamiltonian and its earlier ground are not silently replaced.

The actual cube gives a rigorous changed-reference test. Its retained all-number theorem states $H \ge -\sqrt{3}/4$. The old $N = 4$ ground has centered energy $-2 - \sqrt{3}/4 > -4$, while a singly occupied $N = 8$ product state has expectation -4. Moreover, for $N \le 7$, $K \ge -7/2 - \sqrt{3}/4 > -4$, and particle-hole symmetry transfers this bound to $N \ge 9$. Every new global centered ground therefore has exactly eight particles. A product of four exact dimer grounds on a perfect matching improves the variational bound: inter-dimer hopping expectations vanish because each factor has fixed charge. Hence

$$-4 - \frac{\sqrt{3}}{4} \le E_K \le -2 - \frac{\sqrt{17}}{2}, \qquad \inf \operatorname{spec}(K|_{N\neq 8}) - E_K \ge \frac{\sqrt{17}}{2} - \frac{\sqrt{3}}{4} - \frac{3}{2} > \frac{1}{8}.$$

These are exact full-sector bounds, not relativistic rest masses.

We additionally import Lieb's repulsive Hubbard ground-spin theorem, in the form of Theorem 1, printed p. 8, of his 1993 review; see also the 1989 result and its erratum. The actual hidden-frame hopping is real, spin independent and connected, with a balanced bipartition of four plus four sites and strictly positive onsite repulsion. These finite hypotheses are checked from the source; the theorem allows mixed real hopping signs. Together with the separately proved $N = 8$ ground sector, it gives a unique global centered cube ground with hidden spin zero. The review's centered interaction convention differs from K only by the constant 2. No quantitative neutral gap, physical spin dictionary or thermodynamic uniqueness follows here.

The actual cube also has a faithful full-CAR one-site state. Write $K = T + D - 4$ on $N = 8$. The source identity $A^2 = 3I$ gives $\|T\| = \sqrt{3}/2$. If P_0 projects onto the 256 singly occupied

configurations, every nonzero hopping leaves this subspace, so $P_0TP_0 = 0$. The source's signed vertex flips are transitive. Set $p = \omega(1 - P_0) \le \omega(D)$ and $\delta = \sqrt{17}/2 - 2 > 3/50$. The block Cauchy–Schwarz estimate gives

$$|\omega(T)| \le 2\|T\|\sqrt{p(1-p)} + \|T\|p \le 3\|T\|\sqrt{\omega(D)}, \qquad \omega(T) + \omega(D) \le -\delta.$$

Thus the common site doublon expectation $d = \omega(D)/8$ satisfies $d \ge \delta^2/54 > 1/15000$, while $d \le (\|T\| - \delta)/8 < 1/8$. Gauge, hidden-spin and particle-hole invariance determine the complete one-site state and its modular operator:

$$\rho_x = \operatorname{diag}(d, \tfrac{1}{2} - d, \tfrac{1}{2} - d, d), \qquad K_x = -\log(\tfrac{1}{2} - d)I + \log\left(\frac{\frac{1}{2} - d}{d}\right)(N_x - 1)^2.$$

In particular, $\lambda_{\min}(\rho_x) > 1/15000$ and $\|K_x\| < \log 15000$. For every same-site density σ, put $\Delta = \sigma - \rho_x$ and $\rho_\epsilon = \rho_x + \epsilon\Delta$. The full-site entropy first law and remainder are

$$\left.\frac{dS(\rho_\epsilon)}{d\epsilon}\right|_0 = \operatorname{Tr}(\Delta K_x), \qquad 0 \le D(\rho_\epsilon\|\rho_x) \le \frac{7500\epsilon^2}{1-\epsilon}\|\Delta\|_{\mathrm{HS}}^2, \quad 0 \le \epsilon < 1.$$

This finite result uses no fitted ground vector and covers noncommuting variations. It is not an infinite-ground faithfulness theorem or an identification of the modular operator with a geometric charge.

For a dimer with hopping $t > 0$, let $s = \sqrt{U^2 + 16t^2}$. Its unique full-Fock centered ground is a singlet/doublon superposition with energy $-(U + s)/2$. Its one-site density is

$$\rho_{\mathrm{site}} = \operatorname{diag}(d, \tfrac{1}{2} - d, \tfrac{1}{2} - d, d), \qquad d = (1 - U/s)/4 > 0.$$

Thus this changed reference is faithful on the full local CAR algebra. For a fixed local spin, both addition and removal have energies and weights

$$g_\pm = s/2 \pm t, \qquad w_\pm = (1 \mp 4t/s)/4.$$

Each charge branch has total weight $1/2$; zero weights are retained. At the actual bond parameters $U = 1, t = 1/16$, the positive energies are $(2\sqrt{17} \pm 1)/16$, and the signed second moment is $65/256$.

Normalized full centered-ground projectors, with a tracial exterior, are particle-hole, gauge and hidden-spin invariant. Translation averaging and the existing boundary/volume and local-dynamics estimates give a new infinite-volume ground with positive centered GNS energy and exactly one particle per site in expectation. The old unshifted state is not reused. The exact canonical composite $d = (2n_{\mathrm{opposite}} - 1)c$ gives

$$[K, c(f)] = -c(hf) - (U/2)d(f), \qquad \|d(f)\| = \|f\|.$$

Consequently the same all-word compactness construction supplies a centered eight-valley continuum net. Its moment identities are $M_1 = h$, $M_2 = h^2 + U^2/4$; only the corresponding second-moment upper bound is passed to the continuum. Particle-hole symmetry pairs the complementary valleys and equates averaged addition/removal measures. This resolves the earlier first-moment obstruction, without establishing full spacetime CPT or Lorentz covariance.

There is also a stronger microscopic occupation bound. On a positive kinetic band with $\epsilon \ge \delta > 0$, the GNS maps $Cf = c(P_+f)\Omega$, $Df = d(P_+f)\Omega$ obey $LC + C\epsilon = -(U/2)D$. Integrating $e^{-tL}Ce^{-t\epsilon}$ and using $L \ge 0$, $\|D\| \le 1$ gives $\|C\| \le U/(2\delta)$. Translation invariance, arbitrary

measurable band windows and rational thresholds yield $w(k) \le U^2/(2c^2 \sum_j \sin^2 k_j)$, including the negative-band hole estimate. Folding into the node cells and integrating over a circumscribed ball gives

$$\int \frac{d^3k}{(2\pi)^3 \sum_j \sin^2 k_j} \le \frac{\pi\sqrt{3}}{2} < \frac{11}{4}, \qquad D \le \min\{2, 11(U/c)^2/8\}.$$

The same argument on the original uncentered ground uses $\|B\Omega\| \le \sqrt{r}$ and $r \le 1/2$, giving the separate bound $D \le \min\{2, 11(U/c)^2/4\}$. Neither bound is uniform on a normalized packet in a shrinking momentum region. Native preparation of the new ground, surviving continuum interactions, measured particle identities and the common geometric Einstein premises remain further requirements. The exact source checks and proofs are in `model/centered_source_ground`.

The centered bounded-contact limit is a free field theory. The preceding centered construction admits a stronger conclusion. For $c(a) = v/a$, fixed $v > 0$ and $0 \le U(a) \le U_{\max} < \infty$, every retained continuum net of the canonical charged fields is the free eight-valley Weyl vacuum net. This determines the full field hierarchy, not only a two-point approximation. Its particles and holes have $E(p) = v|p|$ and zero rest masses. These modes are not identified with measured species; the bounded-contact branch does not supply the required interacting theory or its massive sectors.

The proof begins with the exact canonical composite $d = (2n_{\text{opposite}} - 1)c$ and the centered occupation bound $\Delta \le 11u^2/8$, where $u = U/c$. At microscopic time s/c, write $F_c(s; f) = \omega(\{\tau_{s/c}(d(f)), d(f)^*\})$, and let F_0 denote the free filled-sea response at normalized hopping speed one. A degree-m CAR polynomial has centered onsite interaction frequencies in $[-m/2, m/2]$. Bernstein's bound and Duhamel give the degree-three dynamical error. For the state error, the cubic/cubic anticommutator has nine single contractions leaving degree-four words; the two cubic/linear terms leave degree-two words. Their local Wick errors are at most $8\sqrt{\Delta}$ and $2\sqrt{\Delta}$. Absolute row and column sums of the free propagator are at most $e^{3|s|}$, and the contracted factor retains the Schur kernel joining the two smearing indices. Consequently, uniformly over all lattice smearings,

$$|F_c(s; f) - F_0(s; f)| \le \left(3u|s| + 312\sqrt{\Delta}\, e^{9|s|}\right)\|f\|^2.$$

No divergent $\|f\|_1^2$ estimate is used.

The free onsite spin covariance is $I_2/2$, so normal ordering gives $d = 2{:}n_{\text{opposite}}c{:}$ with no linear component. It creates three positive-energy excitations, whose total energy is the sum of three values $e(k) = (\sum_j \sin^2 k_j)^{1/2}$. The spin and antisymmetrization bounds give coefficient sixteen per charge branch. Two independent internal momenta must have $e(k) \le \epsilon$; each such set has normalized volume at most ϵ^3. Thus the complete signed free measure obeys

$$\mu_{0,f}([-\epsilon, \epsilon]) \le 32\epsilon^6\|f\|^2.$$

All eight nodes and both positive physical excitation branches are retained. Negative response frequency labels removal, not a negative ground excitation energy.

Gaussian smoothing now bounds the actual centered composite weight in any fixed physical energy window:

$$\mu_{c,f}([-R, R]) \le \eta_c(R)^2\|f\|^2,$$

$$\eta_c(R)^2 \le \min\left\{1, e^{R^2/(2c^2\epsilon^2)}\left(1536\epsilon^6 + \frac{3u}{\epsilon} + 624\sqrt{\Delta}\, e^{81/(2\epsilon^2)}\right)\right\}.$$

First send $a \to 0$ at fixed ϵ, then $\epsilon \to 0$. This gives $\eta_c(R) \to 0$ uniformly in unit smearings. The composite retains its equal-time CAR norm while its ground spectral weight escapes every finite energy interval.

The next steps use both charge branches in the positive ground GNS space. For the annihilation map $Af = c(f)\Omega$ and composite map $Df = d(f)\Omega$, the source equation is $LA + Ah = -(U/2)D$, with conjugate input space understood. On $h \geq \delta > 0$, integration against the two contraction semigroups gives, after splitting output energies at R,

$$\|\mathbf{1}_{[0,R]}(L)A\| \leq \frac{U_{\max}\eta_c(R)}{2\delta}, \qquad \|\mathbf{1}_{(R,\infty)}(L)A\| \leq \frac{U_{\max}}{2(R+\delta)}.$$

The negative-band creator has the same bound. Taking the two limits and using the CAR determines the entire initial-time state as the pure free filled sea. On a general bounded kinetic window $\|h\| \leq P$, Duhamel followed by integration by parts above R gives

$$\|[C_a(f,t) - C_a(e^{ith_a}f, 0)]\Omega_a\| \leq \frac{U_{\max}}{2}\left(|t|\eta_c(R) + \frac{2+|t|P}{R}\right)\|f\|.$$

The adjoint difference satisfies the same estimate. Passing to the continuum therefore gives both vacuum identities for the free Weyl generator $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$.

Vacuum identities alone would not prove a free hierarchy. Here spatial locality supplies the missing implication. The free filled-sea vacuum is cyclic for the initial CAR algebra outside any bounded spatial region. Indeed, a vector in either Weyl energy band orthogonal to all outside test functions would have compact support. Its Fourier transform would be entire, but the band equation at fixed nonzero transverse momentum contains $\sqrt{z^2 + r^2}$. Continuation around one of its simple branch points reverses that square root while the entire vector and polynomial Weyl matrix remain single-valued, forcing the vector to vanish. This proves density of the projected outside test functions in both bands, separately in every valley. Their particle and hole creation vectors span the one-particle space; Wick recursion then spans all finite exterior products.

For compact smooth f, let $A_b(t,f) = C_b(f,t) - C_b(e^{ith_b}f, 0)$. The retained finite propagation cone and free Weyl finite propagation localize both terms inside one bounded enlargement of the support. Graded locality and $A_b(t,f)\Omega = A_b(t,f)^*\Omega = 0$ make both differences vanish on the dense outside-polynomial vacuum vectors. The initial CAR cyclic space therefore reduces every time field, with free action. Full all-time GNS cyclicity makes it the entire Hilbert space. Thus $A_b(t,f) = 0$ as an operator and the complete canonical charged-field hierarchy is free. No Lorentz axiom or external renormalization theorem was assumed in this argument.

This excludes a surviving interaction and positive rest masses on the stated centered bounded-U branch. It does not decide unbounded coupling trajectories, different field scalings, physical species identifications or the common gauge and Einstein requirements. The full proof, exact bound calculators and countercontrols are in `model/continuum_free_limit`.

Other finite interfaces. The full-CAR locality bounds used above control dynamics at fixed lattice spacing; a chiral spatial continuum remains a further limit. For partition expectations E_c and positive rates,

$$L = \sum_c \gamma_c(E_c - I), \qquad \mathrm{Tr}(X^*LX) = -\sum_c \gamma_c\|X - E_cX\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

What the negative controls exclude. Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound $1/2$ for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

Physical target and remaining obligations. The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may

describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$ for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

This paper's hypothesis and place in the tower

This paper is the matter half of layer 5. The gauge half proposes that gauge structure is the transport of overlap data and classifies the compact Lie type of that transport under complete-response premises [6]. The present hypothesis is that stable massive matter is a localized topological defect of the same native overlap transport, whose rest energy is its excess maintenance dynamics over the reference regime, read in the common clock and energy dictionary (Hypothesis 1.1).

What is proved. The finite results are unconditional statements about stated finite objects: the frustration theorem at fixed holonomy [3], the topology of the native carrier complex, the exact decision test on the native two-load repair process, the collision projector identity, the record-cost theorem for the corrected code, and the normalization, affine-family, rate and scale obstructions for the tested mass prescriptions. The one-generation matter decoder, the Higgs channels and the interaction dictionaries are conditional implications of their declared spin lift, exterior module and actions.

Open joins. The executable model must supply a native defect sector with a protecting invariant under its full admitted law, the excess maintenance energy with kinetics and stress, defect transport with creation, annihilation, fusion and decay rules, an interacting chiral $3+1$ description containing massless radiation, and a frozen dictionary compared with the matched spectrum (Section 8).

Refutation. A proof that every invariant of the full admitted law on the declared carrier and refinement class is constant, including its quantum and complete-history sectors, removes the native defect mechanism for that law. A protected sector whose excess maintenance quantity is not one energy with the common signal speed of its propagation rejects the maintenance reading of mass. A frozen defect dictionary that fails the matched spectrum rejects that law and dictionary.

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1 The defect hypothesis

OPH models observations through finite accessible algebras, retained records and local repair. The executable observer construction supplies explicit carriers, source operators, instruments and a native repair law [7, 8], and the observer layer supplies the record and measurement theorems on which a detector reading rests [2]. A particle in this setting is a statement about a continuing observable response: preparations, currents and detector records must exhibit one excitation, with its quantum numbers, propagation and interactions, in a specified state and clock regime. A finite internal basis vector, memory level, channel decay rate or agreement record does not have that meaning by itself, and no fundamental particle object behind the observers' statistics is assumed.

1.1 Statement

Hypothesis 1.1 (Localized defects of native overlap transport)**.** *Stable massive matter is a localized defect of the native overlap and transport structure, in the following sense.*

(i) Core and reference. *There is a reference regime of the native process and a bounded core outside which the records approach that regime.*

(ii) Protecting invariant. *The transport around the core carries a loop holonomy, a nontrivial invariant relative to the reference, that admitted local repairs cannot erase. The obstruction is to trivializing the transport relative to the reference; observers agree on every measured event.*

(iii) Transport. *Moving the core transports the invariant.*

(iv) One law. *Creation, annihilation, fusion and decay follow from the same native law. Freezing selected seams or supplying punctures to protect an assigned label changes the law and does not count.*

(v) Rest energy. *The rest mass is the excess rest energy of the defect over the reference, produced by its excess maintenance dynamics and read in the same clock and energy dictionary as its propagation and interactions. In a justified local relativistic regime this gives* $E_{\mathrm{rest}} = mc^2$ *with one common signal speed* c*, the speed of the massless response of the same construction. No conversion factor is fitted per species.*

(vi) Unstable species. *Unstable particles are metastable or internal excitations of defects, with decay channels and widths derived from the same law.*

(vii) Radiation. *Massless radiation and the gauge response arise from the same construction, as transport or collective excitations of the overlap structure.*

(viii) Identity. *Species, charges, spin and statistics, and families are properties of defect classes and of their transport.*

The hypothesis is the matter clause of the executable-model hypothesis stated above. It specifies a mechanism and its observable consequences. It is not implied by mass–energy equivalence: a real free scalar field has a contractible field target and trivial vacuum topology, yet its one-particle states have relativistic massive dispersion [9]. A dispersion relation alone therefore does not reveal a topological sector. Whether an effective particle is microscopically a defect is a question about the source realization.

The hypothesis names parameters explicitly. Native attempt counts, host runtime, solver counters and modular parameters are distinct from calibrated observer clocks, and repair counts, entropy production and resource use are distinct from energy. The maintenance-to-energy map in clause (v) must be derived, including kinetic normalization and any reference or regulator subtraction. Different schedules or implementations can change resource counts without changing the claimed physical response; an energy must be invariant under such changes.

1.2 Observable criteria

Each clause has a test that a finite or refined execution can pass or fail. The tests are conjunctive: they must hold for one excitation of one continuing source process, with one state, one clock and one set of accessible operations.

Part of the hypothesis	Concrete test
Defect identity	Exhibit an actual native state or observable sector and a boundary charge invariant under the admitted repairs; derive its coefficient group and transport rules.
Persistence and localization	Retain the full law and show that the core persists or has a controlled lifetime, with reference-like response outside it; distinguish active recovery from autonomous protection.
Maintenance and energy	Derive the excess action or energy of the defect relative to the reference using a calibrated local clock, with all kinetic terms from that same process.
Mass and propagation	In a justified local relativistic regime, recover the same rest energy from propagation and dispersion; the rest-mass relation uses the common signal speed, not a conversion fitted per species.
Particle identity	Obtain spin and statistics, charges, current couplings and any family structure on those same defects or excitations, before comparing mass ratios.

A candidate identification names one excitation or sector of one continuing source process before its mass comparison: the reference regime, local preparation, support, defect invariant, admitted repairs, propagation and allowed creation and annihilation channels. The same object must account for currents, detector response, kinetic normalization and its mass observable. Representation labels, memory levels, decay rates and benchmark field poles cannot be pooled into a spectrum by choosing the closest numerical row.

2 What the finite theorems supply

2.1 Frustration at fixed holonomy

The mathematical seed of the hypothesis is the cycle-obstruction theorem of the consensus layer [3]. Let A be an abelian group, $G = (V, E)$ a finite connected graph with oriented edges, $b_e \in A$ an edge offset and $w_e > 0$ an edge weight. The affine overlap equations $x_v - x_u = b_e$ have a global solution exactly when every cycle holonomy $\sum_{e \in C} \varepsilon_C(e) b_e$ vanishes.

Theorem 2.1 (Persistent frustration at fixed holonomy [3])**.** *Define the mismatch cost*

$$\Phi_b(x) = \sum_{e:u \to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e], \qquad x : V \to A.$$

If some cycle holonomy of b is nonzero, then $\min_x \Phi_b(x) > 0$.

Proof. The cost takes values among the finite subset sums of the positive weights, so its minimum is attained. A zero minimum would solve every edge equation, contradicting the cycle criterion. □

Repairs of the vertex assignments x cannot remove this frustration while b is fixed, and it persists under changes of b inside its cohomology class. The theorem lacks four things that the hypothesis needs. It has no update rule for b; a physical update that changes the compatibility data outside its class can remove the obstruction. It does not localize the frustration in a bounded core. It does not move it. Its residual $\min_x \Phi_b$ is a cost of the whole compatibility problem, not an excitation energy above a reference ground state of its own sector.

On a two-dimensional complex the holonomy and its localization are related by an elementary identity. Let K be a finite oriented two-complex, b an A-valued one-cochain and $F = \delta b$ its face curvature, $F(f) = b(\partial f)$.

Lemma 2.2 (Loop holonomy is enclosed curvature)**.** *For every two-chain c, the holonomy of b around ∂c equals $F(c)$. If $H_1(K;A) = 0$, every closed edge loop is such a boundary, so every loop holonomy is the curvature it encloses. If K is a closed oriented surface with coherently oriented faces, then $\sum_f F(f) = 0$. If $b = \delta n$ is an exact difference, then $F = 0$ and every loop holonomy vanishes.*

Proof. The first statement is the definition of the coboundary. A one-cycle with vanishing homology class bounds a two-chain. The sum of all coherently oriented faces of a closed surface has zero boundary. Finally, $\delta\delta n = 0$. □

On a closed surface with vanishing first homology, a nonzero holonomy around a bounded core therefore requires nonzero curvature inside the core, and the total curvature vanishes: cores come with compensating curvature elsewhere. For a nonabelian structure group the holonomy of a based loop is defined up to conjugation, and its conjugacy class is the invariant that a defect charge would carry.

2.2 A quantitative frustration instance

Frustration produces a strictly positive gap on the executable carrier. Let D_σ be the sign-twisted seam derivative on the connected observer-visible finite local domain, and let $H_\sigma = D_\sigma^* D_\sigma$ use the declared unit-counting measure.

Proposition 2.3 (Exact finite source-Hamiltonian gap)**.** *The signed-incidence formula makes H_σ integer symmetric and positive semidefinite. A kernel vector would supply a sign-consistent coloring on each connected component. The declared reversing seam transport frustrates every triangle, so the connected complex has no such coloring and the signed incidence rank is* 8,662*. Therefore* $\ker H_\sigma = 0$*. With maximum degree* 12*, the determinant bound gives*

$$\lambda_{\min}(H_\sigma) \geq 24^{-8661} > 0.$$

A sparse numerical refinement gives $\lambda_{\min} = 0.11753670336118684$ *with relative residual* 1.49×10^{-14}.

The determinant floor follows because the positive integer determinant is at least one and every eigenvalue is at most 24. The decimal is measured, with a residual check and no interval enclosure. This is the $\mathbb{Z}_2$ analogue of Theorem 2.1 with frustration on every face. It is a global gap of a declared operator. It does not select a localized defect, a physical field fiber or a mass.

2.3 The classification target

The gauge layer supplies the structure group in which defect holonomies would take values [6]. Under its complete-response and endogenous-transport premises, a faithful compact commutator-closed twelve-port response on which proper carrier rechartings act by inner automorphisms has the single compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Under the stated faithful-representation and central-kernel hypotheses the global group is $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. A declared spin lift and exterior module give the one-generation matter module with exact Standard Model charges, anomaly cancellation, the common central kernel $\mathbb{Z}_6$ and the three charged-fermion Higgs channels (Section 5). The carrier premises and the closure hypothesis are stated with that classification [6]; the results below use them only through it.

These are the classification target for defect charges. If the hypothesis holds, the charge of a stable defect is a holonomy class in this group and the matter labels of the decoder are properties

of defect classes and their transport. How the representation labels of the module arise from a defect sector is part of the particle-identity join. The finite gauge identification fixes allowed representations; it does not fix their excitation energies.

2.4 Native topology and the finite flux criterion

The native carrier is the oriented boundary complex of an icosahedron: twelve ports, thirty seams and twenty faces, with Euler characteristic two [4]. The full two-dimensional complex has vanishing H_1, while H_2 does not vanish. The seam graph alone has cycle rank $30 - 12 + 1 = 19$; its cycles all bound in the two-complex. A nontrivial bundle needs actual transition data, and by Lemma 2.2 a nonzero loop holonomy needs face curvature.

Native port loads supply none. The load contrast $b_{ab} = n_a - n_b$ on a seam is an exact difference, so every face circulation is zero; the exact receipt checks all twenty faces on every state of the two-load domain of Section 3. This excludes flux in the actual load-gradient map only. It does not exclude other bundles, enlarged states or quantum sectors.

Declared $\mathbb{Z}_6$ link variables with a puncture pair give a separate flux experiment. Six flux classes, realized by two-puncture witnesses, are distinguished by the spectra of the twisted seam-adjacency operators up to the charge-conjugation pairing $k \leftrightarrow 6 - k$; regauging by a coboundary leaves each spectrum fixed, flux classes add under chain addition, and one edge-midpoint refinement to 42 ports, 120 seams and 80 faces reproduces the six classes. The links and punctures are supplied state and update premises. Unrestricted local link changes can erase the pair. The flux criterion therefore shows what a protected sector would look like on this carrier without exhibiting one in the native process.

3 The native decision test

The simplest local candidate in the native process is a double occupation of one port in the unchanged total-load-two C/E repair process on the twelve ports. Its domain has 78 configurations: twelve double loads and 66 binary states with two unit loads. The test retains all 720 raw outcome maps and compares the same state transitions, transport and cost observables. The invariant test uses the support of all native seams. Lifetime and activity numbers use an explicitly supplied independent uniform-seam and fair-coin probe schedule, which is not the production feedback policy. No physical duration is assigned to an attempt.

Test under the unchanged native law	Exact finite result	Consequence
Configuration sectors	Twelve transient double-load classes feed one closed irreducible binary class of 66 states	A scalar configuration label preserved by every admitted branch is constant on the 78-state domain
Local double-load persistence	Under independent uniform choice of the 30 seams, survival is $(5/6)^n$ and the mean lifetime is six seam attempts	The candidate splits into two unit loads; it is not a protected defect transported by this law
Quantum collision sector	Both complete C/E instruments satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$, with no binary-to-collision feed	The same survival law holds for arbitrary quantum inputs and untouched references
Continuing binary transport	Nonzero current probability 5/33 per attempt in the uniform binary stationary law; every binary event has zero drop of $\Phi = \frac{1}{2}\sum_p n_p^2$	Repair activity and motion are not potential decrease
Proposed edge cost	The binary edge-disagreement cost takes the values 8 and 10, with native transitions in both directions	It is not a monotone maintenance-energy law of this process
Changed-law control	Removing the five seams at one port gives binary closed classes of sizes 11 and 55	A frozen port creates sectors by changing the allowed law

Several entries have short explanations. The native seam rule sets the two endpoint loads to the floor and ceiling of their mean, with a fair coin for an odd remainder. A double load next to an empty port therefore splits whenever one of its five incident seams is chosen, which happens with probability 5/30 per attempt at each of the twelve locations; it never hops as a unit. Two unit loads on the icosahedron disagree across ten seams when they are not adjacent and across eight when they are, and native moves change adjacency in both directions. At a binary seam the rule either waits or swaps, so Φ never changes while loads keep moving. Holding one port leaves either that port loaded with the other load free on eleven ports, or that port empty with $\binom{11}{2} = 55$ placements.

Proposition 3.1 (Collision decay for quantum inputs)**.** *Let $\{K_h\}$ be the complete uniform C or E instrument on the 78-state domain, $K_h = M_h/\sqrt{60}$ over the thirty seams, two coins and retained raw outcomes, including zero-probability outcomes. Then*

$$\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$$

for the projector P_{coll} onto the twelve double-load configurations. For every state ρ on the domain and an untouched reference,

$$\mathrm{Tr}\big[(P_{\mathrm{coll}} \otimes I)(\Phi^n \otimes \mathrm{id})(\rho)\big] = \Big(\frac{5}{6}\Big)^n \mathrm{Tr}\big[(P_{\mathrm{coll}} \otimes I)\rho\big],$$

where Φ is the unread one-attempt channel. No persistent quantum subspace is supported wholly on the collisions under this unread evolution.

Proof. The Heisenberg identity is verified exactly on all retained branches; its effect matrix is diagonal with entries 50/60 on collisions, and no branch feeds a binary state into a collision. The dual channel is linear, so $\Phi^{*n}(P_{\mathrm{coll}}) = (5/6)^n P_{\mathrm{coll}}$, and tensoring with the identity on the reference preserves the identity. A persistent collision-supported subspace would keep its collision weight equal to one. □

These results reject the proposed autonomous configuration defect and the tested automatic cost-to-energy identifications: attempt counts, the quadratic load potential and the edge-disagreement cost. They reject quantum persistence wholly within the collision subspace. They do not exclude other quantum, complete-history or federated defects, another repair law, a common energy offset, or the general hypothesis. A replacement candidate must identify its actual source sector and allowed dynamics before any mass comparison.

3.1 The actively corrected code

The actively corrected matching code is a different persistent object. It uses six fixed nonedge matching words; coherent recovery preserves its logical information, and actual port rotations act on it. Without recovery, one uniform-seam C or E attempt takes the normalized code projector out of the code with probability 31/108 or 55/108, respectively. This is a one-step escape from that preparation, not an all-state lifetime. With the declared recorded recovery, the logical state and its references are preserved and the complete record law has the product form

$$\rho \longmapsto \rho \otimes p(\text{records}).$$

Proposition 3.2 (Record costs cannot split logical states)**.** *For a channel of this product form and any observable c of the records alone, the expectation* $\mathrm{Tr}[(I \otimes c)(\rho \otimes p)] = \mathrm{Tr}(\rho)\,\mathbb{E}_p[c]$ *is independent of the logical state ρ.*

Proof. The trace factorizes over the tensor product. □

The exact receipt checks all 720 raw branches. The one-port cost, each native edge cost and the quadratic load potential compress on the code to $I_6/6$, $I_6/3$ and I_6. A fixed cost of seams or records alone therefore cannot give different masses to different logical states. A common energy offset is not excluded, and no physical offset is derived. Persistence here is active recovery selected by existing records, not autonomous protection, and the code carries no exhibited topological charge. Its fixed matching table and the separately derived equivariant code table must not be identified for a nonscalar observable or a new interaction.

4 Mass from maintenance

4.1 Requirements

A rest mass can be the excess rest energy of a localized excitation over the reference state in a regime where energy, clock and spatial dispersion are defined. To identify it with repair maintenance, the source must derive how the excitation changes an intrinsic action or energy readout, or the same clocked response; subtract the reference contribution; and account for cutoff, volume and renormalization dependence. Propagation and interactions are then tested with that same identification. Attenuation can govern lifetime or decoherence instead of inertial mass, and the response calculation must decide which.

Counting alone cannot supply the scale.

Proposition 4.1 (Count-rate no-go)**.** *An oriented register with twenty-four slots, a twenty-four-stage automaton, or a decomposition into twenty-four formal update steps determines no physical frequency, Hamiltonian gap or mass scale.*

Proof. Rescaling every physical generator by $H \mapsto \lambda H$, $\lambda > 0$, preserves the register, transition graph, grading and settled quotient while multiplying all generator gaps and rates by λ. □

The number twenty-four plays no role in the argument, and a repair iteration counter is likewise not an operational clock. A physical rate requires a source-derived clock instrument, a calibrated comparison line and a residual bound. Once such a line exists, a scale follows exactly.

Theorem 4.2 (Conditional determinant-clock attachment)**.** *Let L_{24} be a one-dimensional source-derived operational clock line with a distinguished calibrated positive gap vector ℓ_{24} of norm Δ_{24}. Suppose there is a refinement-natural, quotient-visible norm-preserving line isomorphism $\Theta_e : L_{24}^{\otimes 3} \to \det M_e$ to the physical charged-lepton determinant line, with the element compatibility $\Theta_e(\ell_{24}^{\otimes 3}) = \det(M_e)$. Then $g_e := |\det M_e|^{1/3} = \Delta_{24}$. If the element-compatible map rescales norms by a separately derived factor $\kappa > 0$, then $g_e = \kappa^{1/3}\Delta_{24}$.*

Proof. The distinguished tensor-cube vector has norm Δ_{24}^3. Element compatibility and norm preservation give $|\det M_e| = \Delta_{24}^3$. □

Writing a norm-preserving line isomorphism without the distinguished-vector equality constructs neither the clock line nor the attachment. Data that fix only centered logarithms leave the scale free.

Theorem 4.3 (Absolute-scale underdetermination)**.** *Any readout factoring only through a centered log triple (ℓ_1, ℓ_2, ℓ_3), $\sum_i \ell_i = 0$, and its difference invariants leaves the common scale $g > 0$ of $m_i = g\, e^{\ell_i}$ free.*

Proof. For every $c \in \mathbb{R}$, $e^c \operatorname{diag}(e^{\ell_i})$ has the same centered logarithms and differences, determinant e^{3c}, and masses scaled by e^c. □

A rest energy appears to a comoving observer as a clock rate. On a declared Lorentz module and mass shell, free plane waves persist under the declared translation flow, and their phase along their own frame has proper-time derivative m. For $m > 0$ the internal-clock periods are $(2\pi/m)\mathbb{Z}$; the zero-mass clock is constant and has no fundamental period. This is conditional kinematics, without a source-selected mass or clock, and it states the target that a maintenance rate must meet: the same m must appear in the internal clock, in the dispersion and in the kinetic normalization.

Kinetic and mass terms must come from the same source. For a four-component Dirac convention with positive kinetic metrics,

$$Y_{s,\mathrm{can}} = Z_L^{-1/2} Y_s Z_R^{-1/2} / \sqrt{Z_H};$$

a literal all-left Weyl bilinear has a transpose on the left normalization factor. A record matrix can change a coordinate without changing the mass: $Z_L = Z_R = R > 0$ and $Y_s = g_s R$ give $Y_{s,\mathrm{can}} = g_s I/\sqrt{Z_H}$ and $\det(pR - hg_sR) = \det(R)(p - hg_s)^3$. A hierarchical Gram matrix by itself need not produce a hierarchical physical vertex. Eliminating an invertible heavy block also changes the Higgs derivative,

$$\partial_H(A - BD^{-1}C) = A_H - B_H D^{-1}C - BD^{-1}C_H + BD^{-1}D_H D^{-1}C.$$

Repair contraction and coherent propagation are different dynamics. Passive stationary fixed-reference schedules have an exact rank-two Schur response obstruction, including eight zero crossing weights among the 22 seams, and positive whitening preserves the null direction. The uncorrected recorded 66-state channel has no invariant fifteen-state work code in the tested class. Its ten-mode response conditioned on an incident event has operator norm $\lambda_C = 7/10$ or $\lambda_E = 2/5$.

Proposition 4.4 (Contraction has no direct unitary intertwiner)**.** *Let R have operator norm $\lambda < 1$, let U be unitary and let J be any linear map. Then $\|UJ - JR\|_{\mathrm{HS}} \geq (1-\lambda)\|J\|_{\mathrm{HS}}$. In particular, the conditioned native response has no nonzero exact direct intertwiner to the unmeasured closed quadratic Weyl evolution.*

Proof. $\|UJ\|_{\mathrm{HS}} = \|J\|_{\mathrm{HS}}$ and $\|JR\|_{\mathrm{HS}} \leq \lambda\|J\|_{\mathrm{HS}}$; apply the triangle inequality. □

The proposition does not exclude a field coupled to detectors, an environment or interactions; a complete dephasing-channel control realizes the same uniform contraction. A contraction modulus describes attenuation in its declared event parameter, and the argument of a transfer eigenvalue describes a phase only with an unwrapping and clock convention. Neither becomes a rest mass by taking a logarithm.

4.2 Obstructions for the tested mass prescriptions

The following results are theorems about specified classes of uncalibrated prescriptions. Each excludes its class as the charged mass law. None excludes other repair laws on the carrier.

The complete record memory. A complete twelve-state family memory uses the native events with probabilities $p = 1/12$ and $q = 1/6$, a specified reset state and declared half-turn updates. All 128 native raw labels are combined with all twelve old memory states; conditioning on successful events is not substituted for this 1,536-branch law. The stationary projector average

$$Y = \sum_k \pi_k P_k \succeq 0, \qquad \operatorname{tr} Y = 1,$$

has coefficient ratios $1 : 195.11703757 : 3579.82624571$. A two-angle record inverse cancels the finite-pulse bias exactly; the resulting numerical allowance is conditional on its floating-point and mixing estimates. Gating the same memory with the complete thirteen-outcome matter instrument

$$M_0 = \cos(t\sqrt{C}), \qquad M_a = iF_a \frac{\sin(t\sqrt{C})}{\sqrt{C}}, \quad a = 1, \ldots, 12,$$

with $C = \sum_a F_a^\dagger F_a$ built from the matter lift of the trace-orthonormal current frame, and with both components of each Higgs tensor, gives sector-dependent coefficients without empirical mass inputs. At the declared $t = \pi/5$ the ratios for the channels matched after construction are $1 : 271.446 : 6071.036$ (up), $1 : 311.762 : 6822.880$ (down) and $1 : 675.782 : 30775.333$ (charged lepton). The entire declared catalog $t \in \{\pi/5, \pi/3, \pi/2\}$ fails the charged hierarchy comparison; its zero coefficient at $t = \pi/2$ is retained. This finite candidate uses supplied control services and a declared architecture. It is a tested prescription, not a derivation of the spectrum.

Proposition 4.5 (Normalized-feedback scale bound)**.** *For a probability average of rank-one projectors on a three-dimensional family space, $Y \succeq 0$, $\operatorname{tr} Y = 1$ and $1/3 \leq \lambda_{\max}(Y) \leq 1$. With common canonical kinetics, unit-modulus neutral Higgs tensors and a common Higgs norm v, the three family action masses in each sector sum to v. Any two sector maxima have ratio at most three.*

Proof. The nonnegative eigenvalues sum to one, so their largest lies between their mean and their sum. The specified Higgs tensors multiply each family singular value by v. Apply the bounds to both sectors. □

This is a sector-normalization obstruction under the common-coupling and canonical-kinetic hypotheses. It does not bound a smallest eigenvalue away from zero and does not exclude within-sector hierarchies. A sector-dependent strength or a different kinetic response changes the hypotheses and needs its own derivation. Positive trace-one matrices can differ in eigenbasis, so the bound implies nothing about mixing or CP violation. Unconditional event-duty pulses change the normalization and must use the complete correlated record law: for the reset projector P_2 the C-duty response is $Z_C = Y - (1-p)P_2$, not pY. The always, C, E and CE policies have traces $1, p, q, pq$; their sector-maximum spans are approximately $1.0233, 2.0572, 1.1528, 2.3695$, and the E and CE responses have rank two. These policies are excluded as the charged mass prescription.

Static family features. The native quadratic comparison uses $Y_u = aF$, $Y_d = bF$, $Y_e = JF$ with

$$a = \log(25/8), \quad b = \log(8/5), \quad J = a - b, \quad v = a + b.$$

The family matrix F has eigenvalues

$$\frac{61}{450}, \qquad \frac{41}{225} \pm \frac{7\sqrt{5}}{750},$$

and the direct positive prescription has ratios approximately $1 : 1.1903 : 1.4982$, far from a charged hierarchy. Complex affine coefficients do not repair the class.

Proposition 4.6 (Affine quadratic-family exclusion)**.** *Let $Y = uI + wF$, $u, w \in \mathbb{C}$, with singular values $m_{\min} \le m_{\mathrm{mid}} \le m_{\max}$. With $\varphi = (1+\sqrt{5})/2$,*

$$m_{\max} \le \varphi^2 m_{\mathrm{mid}} + \varphi m_{\min}.$$

Proof. For ordered eigenvalues $f_1 < f_2 < f_3$ of F, the complex points $z_i = u + wf_i$ are collinear, and $f_3 - f_1 = \varphi^2(f_2 - f_1)$ gives $z_3 - \varphi^2 z_2 + \varphi z_1 = 0$. Solving this relation for each point and applying the triangle inequality bounds any modulus by φ^2 times one of the others plus φ times the third. The middle point is a convex combination of the endpoints. Applying these bounds to the largest modulus gives the sorted inequality. □

Conditional on $m_{\mathrm{mid}}/m_{\min} \le 207$, the bound gives $m_{\max}/m_{\min} \le 543.551$, whereas the broad charged-lepton pole-ratio windows $[206, 207]$ and $[3470, 3485]$ require at least 3470. The adopted $\overline{\mathrm{MS}}$ comparison at M_Z has lepton ratios about 210.7 and 3578 [13], and the inequality excludes that comparison as well. The bound is not a universal heavy-to-light bound: a light eigenvalue near zero can make that single ratio arbitrarily large. The class has a fixed quadratic action, so its anticommutator poles are state independent; an occupation-versus-gap factor of two, a common clock rescaling or a basis change cannot change their ratios.

Other explicit laws. A five-memory feedback law gives ratios near $1 : 30.84 : 452.62$; a two-rotation memory gives $1 : 382.23 : 5909.43$; the admitted auxiliary-dressing class bounds its two smallest subthreshold lepton poles by a ratio below 32. A near numerical ratio found by searching many prescriptions is not a blind prediction; the selection ancestry and all admitted alternatives must be retained.

Symmetry and entropy selectors. Residual symmetry does not select the ratios. Realize the quadrupole carrier W_5 as traceless symmetric real 3×3 matrices. Invariance under a threefold or fivefold rotation about n forces $A = \alpha(nn^T - I/3)$, with a double eigenvalue. The twofold fixed locus is three-dimensional and admits simple spectra, and every subgroup of A_5 other than the trivial group, C_2 and V_4 contains a threefold or fivefold rotation. No residual stabilizer pins an isolated simple-spectrum orbit. For six antipodal pair weights $p_i = (1 + ra_i)/6$ at fixed index of coincidence, every global minimizer of the full uniform-reference relative entropy $D(p\|\mathbf{1}/6)$ for $0 < r \leq \sqrt{60}$ is a permutation of one high and five equal low weights, the standard maximum-entropy branch at fixed index of coincidence [12]. Its quadrupole spectrum is $(-2/\sqrt{15}, -2/\sqrt{15}, 4/\sqrt{15})$ up to scale, with the fivefold-axis stabilizer. No amplitude rule therefore yields three distinct masses through an attachment that reads only these eigenvalues. The leading cubic truncation selects the same degenerate orbit. The strict full-support minima of the quartic truncation have sorted-gap ratio φ or a double eigenvalue, and the quartic truncation reaches the central charged-lepton gap ratio only at a target-attached amplitude on its closed boundary, where eight of twelve port probabilities vanish; that is not a prediction. These exclusions concern the stated reference, constraint and readout.

4.3 Calibrated control and target-informed constructions

A calibration packet supplies nine running charged Yukawas, mixing parameters and the Higgs scale in stated schemes. The finite action checks its internal mass and current identities, and separate source-record experiments and a moving-field readout recover the supplied running parameters with a fixed probe design and a blind inverse. This is a readout test: the Hamiltonian retains the empirical inputs, and decoder blindness does not remove that ancestry. A running action parameter is not automatically a laboratory pole mass.

Several numerical constructions have target ancestry and are retained only as such: fixed closure-coordinate running charts for quarks, the electroweak bosons and Higgs criticality; an eight-register charged-lepton response architecture; a down-type Clebsch assignment ranked against measurements; and the equal-weight block realization of the Koide balance. They are conditional constructions: their discrete choices and some attachments have target ancestry. Freezing them does not erase that ancestry. Their transmutation recipe uses supersymmetric one-loop coefficients $(33/5, 1, -3)$, whereas the stated matter content of three fifteen-state generations and one Higgs doublet gives $(41/10, -19/6, -7)$ in the same normalization, and their energy unit is the Planck energy formed with measured G. A square of renormalization and Higgs maps that returns the same declared tuple along both routes is a declared-map packaging identity, not a construction of source-derived comparison maps. The current packet supplies no OPH result about Higgs naturalness. None of these constructions is a derivation, and the measured particle spectrum is not derived.

5 Matter and interaction readout from the finite response

The finite response determines more than the abstract gauge Lie type. Its matrix generators support an exact, conditional matter decoder and local interaction tensors. A representation is distinct from a physical particle: a particle identification also consumes its spin, statistics, propagator, interaction and observer-clock interpretation.

5.1 From response operators to a fifteen-state matter module

Let a response generator on the six-dimensional carrier have blocks $D = \operatorname{diag}(A, B)$, with $A \in \mathfrak{u}(3)$ and $B \in \mathfrak{so}(3)$. Write $B = \widehat{w}$. The declared spin lift and central trace redistribution give

$$L(D) = \operatorname{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\operatorname{tr} A}{2} I_2\right).$$

This is a Lie-algebra homomorphism. The complete twelve-direction response and all sixty-six bracket pairs are checked exactly. The central direction, after clearing denominators and choosing the weak-positive orientation, has primitive integer character

$$q = \operatorname{diag}(-2, -2, -2, 3, 3).$$

The lift and trace redistribution are hypotheses. A native weak rotation through 2π is the identity on its vector carrier and acts as minus the identity on the lifted doublet. Thus the lifted module is not a subrepresentation of the original six-dimensional response. Global descent and a physical current interpretation require the corresponding additional identifications.

Proposition 5.1 (Conditional matter decoder)**.** *For the lifted five-dimensional one-body action, the representation*

$$M = \Lambda^2 \mathbb{C}^5 \oplus \Lambda^4 \mathbb{C}^5$$

has dimension fifteen. Its charge and quadratic Casimir operators identify the following irreducible components without a field-name input table.

Component	Color	Weak dimension	$Y = q/6$	Dimension
Q	3	2	$1/6$	6
u^c	$\overline{3}$	1	$-2/3$	3
d^c	$\overline{3}$	1	$1/3$	3
L	1	2	$-1/2$	2
e^c	1	1	1	1

Proof. Split $\mathbb{C}^5 = V_c \oplus V_w$, of dimensions three and two and charges -2 and 3. Then

$$\Lambda^2(V_c \oplus V_w) = \Lambda^2 V_c \oplus (V_c \otimes V_w) \oplus \Lambda^2 V_w,$$

with integer charges $-4, 1, 6$. The degree-four summands are $\Lambda^3 V_c \otimes V_w$ and $\Lambda^2 V_c \otimes \Lambda^2 V_w$, of charges $-3, 2$. Color and weak Casimirs separate the representations, and Cartan weights distinguish the conjugate color character. Exact projectors give the same answer after a rational change of basis mixing unlike charge sectors. □

The auxiliary exterior construction has two additional invariant lines, the vacuum and top exterior degree. Removing them is part of the declared fifteen-state module. Charge traces alone do not remove neutral invariant states. Its even auxiliary parity also does not establish physical fermion statistics. Physical two-component Weyl fields require an independent CAR algebra and spin interpretation. The anomaly-free charge table and common central kernel are conditional representation results; they do not identify laboratory species or select a generation count.

5.2 One Higgs doublet and the exact interaction tensors

For two same-chirality physical Weyl fields, the antisymmetric spinor contraction makes the internal coefficient symmetric. The Lorentz-scalar internal pair space is consequently $\mathrm{Sym}^2 M$, of dimension 120. A color-Casimir kernel and weak-Casimir projector identify one doublet of hypercharge $-1/2$ and two of hypercharge $+1/2$. Their constituent integer-charge pairs are $(-4,1)$, $(-3,6)$, and $(1,2)$. These are interpolating channels; their existence supplies no bound-state pole, condensate or elementary-boson commutation relation.

Proposition 5.2 (Conditional one-doublet interaction)**.** *Declare one scalar doublet H of hypercharge $1/2$. Each of the three channels above has a one-dimensional invariant contraction space. They give*

$$Qu^cH, \qquad Qd^cH^*, \qquad Le^cH^*.$$

The same scalar doublet and its conjugate occur in all three expressions.

Proof. Restrict the exact pair generators to each color-singlet doublet. The tensor sum of that doublet generator and the generator of H, or its conjugate, has a one-dimensional common kernel. Equivalently, its coefficient matrices C_α satisfy, for each of the twelve source currents,

$$T^tC_\alpha + C_\alpha T + \sum_\beta (T_H)_{\beta\alpha} C_\beta = 0.$$

Here the conjugate action is $-T_H^t$ in the skew-Hermitian convention. The physical pairing kernel $\epsilon \otimes C_\alpha$ is antisymmetric. The identity $S^t\epsilon S = \epsilon$ for $S \in \mathrm{SL}(2,\mathbb{C})$ establishes its Lorentz scalar character. The executable checks the rotation and boost generators as well as all current identities. □

For three declared families the coefficients are three complex 3×3 matrices, with 27 complex entries before flavor-basis redefinitions. The pairing convention is

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{2}\epsilon_{AB}\psi_A^t K(H)\psi_B + \mathrm{h.c.}$$

up to an overall sign. With canonical kinetic normalization and $H = (0, v)$, its singular values are tree-level mass parameters; $v = v_{\mathrm{SM}}/\sqrt{2}$ in the conventional electroweak normalization. Exact matrix-unit tests cover arbitrary complex family coefficients. Their values, the Higgs potential and its vacuum are not selected by these identities. The optional invariant $(LH)(LH)/\Lambda$ has six complex symmetric family coefficients and gives $M_\nu = \kappa v^2/\Lambda$. It is an additional dimension-five effective interaction, with no selected coefficient, scale or ultraviolet completion.

If both family factors are identified with the same unbroken irreducible rank-three source representation and the Higgs is family-neutral, the exact commutant has dimension one. In that branch each Yukawa intertwiner is yI_3, giving equal singular values. A nonscalar background can split the band; choosing that background is an additional physical hypothesis. The source family-band theorem therefore does not imply the observed hierarchy.

5.3 Scope of the readout

A declared coherent filter on the native six-dimensional response admits an exact invertible twelve-control map into the same compact algebra; the spin/exterior and Higgs tensor maps above apply after this change of controls. It must not be identified with state cooling. For a rank-three projector P, the cooling channel $\Phi(\rho) = P\rho P + \mathrm{tr}((I-P)\rho)P/3$ has dual $\Phi^*(B) = PBP + \mathrm{tr}(PB)(I-P)/3$,

which is not the Hamiltonian map $h \mapsto PhP$. Its classical recorded branches lose the relative phase of the two triplets.

The executable calculations reconstruct the response, representation projectors and interaction tensors from repository-local source operators, with exact rational or algebraic identities separated from numerical residuals. They use no measured particle masses. They supply neither a physical chiral field net nor a source-selected Higgs vacuum, Yukawa matrix or mass pole. A relaxation gap, finite cavity frequency or commuting internal modular offset requires its own mass-shell identification before it can be called a physical rest mass.

6 Further supporting results

6.1 Currents and carrier covariance

The native protected geometric object has type $3 \oplus 3'$. Twelve ports count interfaces, not independent Lie generators. The tested native exchange response closes at seventeen currents. A specified law instead closes at $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$, of dimension twelve, and a coherent Hamiltonian filter has an exact invertible control map into the same algebra. This is a conditional control construction; ordinary C/E state repair is not shown to select the physical gauge response. The full exchange-plus-class-sum controller has eighteen directions; a fixed-bias long-time limit has six observable directions, and a joint bias and update limit with bounded drive approaches twelve.

The carrier recharting premise acts by inner automorphisms on the current algebra, and the real part of the colour generators implements rotations about the port axes. These facts establish carrier covariance. They do not establish which transformations observers recover as physical spatial rotations: equivalent finite-group representations on different factors need not be one physical action, and declaring one action to be both frame rotation and colour transport does not establish their independence. A rank-three local carrier frame is not a proof of three-dimensional physical propagation on a federation.

6.2 Finite interaction identities

With a declared scalar quartic, breaking scale and gauge couplings, the finite electroweak-breaking theorem identifies the vacuum shell, its one-dimensional stabilizer, three gauge-orbit Hessian zero directions and the tree gauge-mass ratio. The slot dictionaries then give photon charge eQ, neutral shape $g_Z(T_3 - Q\sin^2\theta_W)$, and charged shifts $g_W/\sqrt{2}$ on doublets and zero on singlets. These are identities of the declared infinitesimal action and pairing, with supplied constants; they construct no physical current operators, mixing, decay amplitudes or Fermi limit.

For a declared complex algebra with $[X_i, X_j] = iq\epsilon_{ijk}B_k$, the Pauli identity reads

$$(\sigma \cdot X)^2 = (X \cdot X)I - q\,\sigma \cdot B,$$

which in the stated orbital and spin normalization is $g = 2$. It constructs no gauge potential, electron, anomalous moment or physical coupling.

The ten-row exterior census follows from the 32 subsets of five modes: bidegrees (c, w) have multiplicity $\binom{3}{c}\binom{2}{w}$ and charge $-2c+3w$, and removing vacuum and top degree leaves ten bidegrees. A supplied anomaly-free chiral mask and central weights recover the same six-element kernel. The finite family interface selects rank three under its single-object, faithful-action and operational-cost hypotheses; a separate conditional argument from intrinsic CP capability and weak-sector ultraviolet completeness gives $3 \le N_g \le 5$. Neither attaches a physical family field or excludes extra light sectors.

On the declared carrier $C \oplus W = \mathbb{C}^3 \oplus \mathbb{C}^2$, with $H = h_0 e_3 + h_1 e_4$, the top-form contractions $B_u = [Q \wedge H \wedge u^c]_5$ and $B_d = [(\iota_{\overline{H}} Q) \wedge d^c]_5$ give

$$U(H) = I_C \otimes \begin{pmatrix} h_1 \\ -h_0 \end{pmatrix}, \qquad D(H) = I_C \otimes \begin{pmatrix} -\overline{h}_0 \\ -\overline{h}_1 \end{pmatrix}, \qquad U^\dagger U = D^\dagger D = \|H\|^2 I_C, \quad U^\dagger D = 0.$$

The full doublet has unit-magnitude color Clebsches, and the channel coefficients g_u, g_d are not fixed by normalizing these tensors. On a family attachment $F \otimes M$ that restricts every supplied operator to $I_F \otimes A$, the result is $Y_u = g_u I_F$ and $Y_d = g_d I_F$, preserved by products, adjoints, inverses, positive functional calculus and family-independent heavy-mode elimination. This bounded direct-contraction model produces no hierarchy.

6.3 Flavor constraints

Species are identified by represented charges, Casimirs and interaction channels. Sorting a nondegenerate singular spectrum names mass eigenstates; it does not derive three families or their weak couplings. For canonical Dirac matrices with $U_{sL}^\dagger Y_s U_{sR} = \mathrm{diag}(y_s)$, the charged-current matrix is $V = U_{uL}^\dagger U_{dL}$, and a physical CP test uses a rephasing-invariant quantity such as the Jarlskog invariant [54, 15]. Yukawa matrices simultaneously real in one weak basis have zero CP-odd invariant, and a common matrix up to sector scalars has trivial charged-current mixing for a simple spectrum. Orientation reversal of a cycle phase is physical time reversal only with its own antiunitary intertwiner, and a classical cycle affinity is a log probability ratio, not a coherent U(1) phase.

For a positive Hermitian circulant square-root response $C = aI + bR + \overline{b}R^2$, $R^3 = I$, $b = are^{i\delta}$, with eigenvalues the nonnegative square roots of the masses,

$$Q := \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{1}{3} + \frac{2}{3} r^2,$$

since $\mathrm{tr}\, C = 3a$ and $\mathrm{tr}\, C^2 = 3a^2 + 6|b|^2$. Thus $Q = 2/3$ exactly when $r = 1/\sqrt{2}$ [10, 11]. The identity selects no positive branch, phase, absolute scale or physical response; the balance condition that realizes $Q = 2/3$ is a premise with measured-triple ancestry.

Conditional on the registered source signature containing no typed datum that separates the stated centered-spread orbit, a generic quark package does not define a unique quark mass spectrum. Granting an ordered profile shape, write its centered logarithms as $\sigma_u \widehat{\ell}_u$ and $\sigma_d \widehat{\ell}_d$. Independent positive changes of (σ_u, σ_d) preserve the certified gauge representations, anomaly traces and invariant Higgs lines, so those certificates leave at least an $(\mathbb{R}_{>0})^2$ fiber in this granted subfamily. Without the shape hypothesis, sector means and shapes are also undetermined. Representative independence does not select the datum: a constant map on a bridge fiber descends to its quotient whatever its value, and a target-derived spread keeps that ancestry after descent. Two finite exclusions complete the picture. The candidate reciprocal-ray law requires a product of adjacent log-gap ratios equal to one, whereas common-scale $\overline{\mathrm{MS}}$ Yukawas at M_Z give approximately 0.83532 [13]. Directly identifying two real A_5 residual axes with the Cabibbo angle fails: the least nonzero angle is about 20.905°, with sine 0.356822, against the measured approximately 0.225. Under additional alphabet, measure-balance and register-faithfulness hypotheses a finite enumeration selects the Clebsch set $\{1/3, 1, 3\}$. Ranking its six generation assignments against measurements is target-informed. The selected assignment $y_b = y_\tau$, $y_s = y_\mu/3$, $y_d = 3y_e$ is the Georgi–Jarlskog texture [46], and it fails the adopted joint lattice comparison [14].

6.4 Response readout tools

The response diagnostics test a field interpretation once a defect sector exists. The finite interacting CAR response determines its inverse and frequency derivative from all Lehmann residues, with canonical sum-rule and positivity admission. Kinetic and mass forms transform together under a change of interpolators, so raw matrix eigenvalues are not invariant mass data. A charge-resolved census of the supplied 180-mode Weyl–Nambu symbol retains signed and dark modes and unresolved frequency lists.

For one charged L/e^c pair, a finite neutral-reference experiment reconstructs the complex two-mode transfer from complete detector and clock records. Charge-compatible source pulses prepare a fixed-total-charge-zero reference, and all measured continuation branches are kept. On a supplied principal logarithm window and canonical quadratic response class, the inverse recovers the generator, residues and joint z-axis momentum and neutral-Higgs multiplier derivatives. The retained 304 contexts have 4,864 terminal outcomes. Withheld settings and a second duration give generator residuals at most about 6.40×10^{-8} per unit of reconstructed clock coordinate and complete-output distances about 4.36×10^{-8}; both durations repeat one elementary source word, with composition defect about 2.23×10^{-16}. These are numerical diagnostics. Phase-sensitive controls detect opposite coupling signs hidden by Rabi marginals, and withheld momenta detect an otherwise indistinguishable two-hop kinetic control. The experiment reconstructs a component of a supplied action; it selects no mass coefficient and introduces no particle candidate.

A finite abelian quantum link has an electric–matter interaction and an exact operator-Gauss sector on a restricted four-state code. A Euclidean overlap calculation on the fifteen-state representation checks the modified chiral projector and gauge covariance on a nonconstant nonabelian background, with a stated small-plaquette locality domain; exact gauge-invariant constructions of this type are known for abelian chiral lattice theories [21, 22, 23]. The free spatial implementation has conditional Weyl–Higgs dispersion on a supplied background; its cubic staggered continuum contains two Dirac tastes and both chiralities [100, 101], and the naive lattice has eight balanced Weyl nodes. Deleting unwanted nodes is not a decoupling proof.

7 One interacting bound pair and its remaining physical joins

The finite construction follows one excitation through protection, motion, charge, energy and a complete spectral readout. It uses the supplied three-dimensional CAR lattice action

$$H = T + UP, \qquad T = d\Gamma(h), \qquad h(k) = -c\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad P = \sum_x n_{x0}n_{x1}, \qquad c = \tfrac{1}{8}, \quad U = 1,$$

on the complete antisymmetric two-fermion sector. Here P is the orthogonal projector onto on-site singlets. The lattice, CAR dictionary, couplings and Hamiltonian model-time convention are inputs. Periodic boxes resolve momentum; the open boxes used for regional observations have the same local terms but different boundaries. Infinite volume below keeps lattice spacing fixed. Repulsively bound lattice pairs are established physics [97]; the contribution is their explicit same-action attachment to the source-observation construction.

Proposition 7.1 (Isolated pair band on the declared lattice)**.** *Let $b = 2\sqrt{3}c$ and $U > b$. On a periodic box with V sites, exactly V eigenvalues of H lie in the upper band, all at least U, while every remaining eigenvalue is at most b. Thus the separation is at least $U - b$. There is one upper-band state in each total-momentum fiber.*

Proof. The symbol obeys $h(k)^2 = c^2 \sum_j \sin^2(k_j) I$, so $\|T\| \le b$. A single hop separates an on-site pair, hence $PTP = 0$. The quadratic form of H on $\operatorname{Ran} P$ is exactly U, while on its orthogonal complement it is that of T and is at most b. Min–max gives the two bounds and the band count. In each momentum fiber P has rank one. □

Writing $s = (0, 1, -1, 0)/\sqrt{2}$ and $d_K(k) = s/\sqrt{V}$, the free fiber is $B_K(k) = h(k) \otimes I + I \otimes h(K-k)$. Its upper pole and contact residue satisfy

$$g(E, K) = \frac{1}{V} \sum_k \langle s, (E - B_K(k))^{-1} s \rangle, \qquad Ug(E, K) = 1,$$

$$Z(K) = \left[\frac{U^2}{V} \sum_k \langle s, (E - B_K(k))^{-2} s \rangle \right]^{-1}, \qquad \psi_K = U \sqrt{Z(K)} (E - T_K)^{-1} d_K.$$

The resolvent is positive above the free spectrum and g is strictly decreasing there. The isolated pole is therefore unique. The receipt keeps the complementary spectral weight $1 - Z$, including all multiplicities and all physical states; it does not condition on contact detection.

The same resolvent gives relative localization. If $\Pi_{\ge R}$ selects relative Manhattan distance at least R, a hop changes that distance by at most one, and the first R terms of the Neumann series cannot contribute. For integers $R \ge 0$,

$$\|\Pi_{\ge R} \psi_K\|^2 \le \min \left\{ 1, \left(\frac{U}{U - b} \right)^2 \left(\frac{b}{U} \right)^{2R} \right\}.$$

This is spectral protection and localization under the closed declared Hamiltonian, not protection against arbitrary native C/E branches. The positive contact overlap fixes a global phase for the band. Continuously reducing c to zero preserves the separating gap and contracts the band to on-site pairs. Its line bundle is topologically trivial; no nontrivial native boundary holonomy is assigned by this construction.

The retained $L = 4$ calculation Fourier-includes all 64 band states into the complete 8,128-dimensional position-space CAR sector. An independently assembled hopping/contact Hamiltonian verifies that inclusion, and its evolution agrees with band propagation. A predeclared wavepacket changes the complete site-contact/separated outcome law by about 0.08986 at model time 16. Every relative and spin component remains present. On-site pair creation $b_x^* = a_{x0}^* a_{x1}^*$ obeys

$$[N, b_x^*] = 2b_x^*, \quad (b_x^*)^2 = 0, \quad [b_x, b_x^*] = 1 - n_{x0} - n_{x1}, \quad [b_x^*, b_y^*] = 0 \quad (x \ne y).$$

These are charge-two, even-parity, hard-core composite operators in the supplied CAR theory. The conserved number current is checked independently on the full finite action. Electromagnetic charge, Standard Model representations, generations and Lorentz spin remain unassigned. The Weyl hopping does not conserve standalone internal spin; the exact singlet statement below concerns its specified momentum.

7.1 One energy difference, one clock corner, and two mass readings

At $K_\eta = (\pi, \pi, \pi)$, $h(K_\eta - k) = h(k)$, and the total-spin operator kills s. Thus $Hd = Ud$ and $Z(K_\eta) = 1$. On each even periodic box of side at least four, $r(k) = \sqrt{2} \sin(k_x) d(k)$ is normalized, orthogonal to d and satisfies the physical exchange constraint. It has $Pr = Tr = 0$. The invariant space $S = \operatorname{span}\{r, d\}$ therefore has $H|_S = \operatorname{diag}(0, U)$. Both levels have number two and the same total momentum; r is not a vacuum.

The source clock maps native occupations $\{1,2\}$ and $\{7,2\}$ to (r,d) by an explicit supplied isometry. Their product-reference weights are $200/7505$ and $64/7505$, so the heralded corner has success mass $264/7505$ and normalized density

$$\rho_S = \operatorname{diag}(25/33, 8/33) = e^{-\beta H|_S}/\operatorname{Tr}(e^{-\beta H|_S}), \qquad \beta U = \log(25/8).$$

Complete source executions retain both herald outcomes, native C/E continuation, leakage, the record-derived clock and energy readout. Their unconditional half-trace error bound is below 0.018762; it is never divided by the herald probability. The Heisenberg model coordinate is $t = -\beta s$ for modular parameter s. Faithfulness is only on M_2; neither the full spatial isometry nor a native full-field Gibbs preparation has been executed, and physical seconds and energy units are not established.

For the same band in infinite volume, let κ be the curvature $\partial^2_{K_j} E$ at K_η. Resolving the singlet/triplet blocks in second-order perturbation theory gives

$$\kappa = \frac{2c^2}{U}\left\langle \cos^2 k_j \left[1 + \frac{4c^2(\sum_l \sin^2 k_l - \sin^2 k_j)}{U^2 - 4c^2 \sum_l \sin^2 k_l}\right]\right\rangle_{\mathrm{BZ}}.$$

Indeed, at K_η the first derivative couples d to the triplet through $c\cos k_j (I \otimes \sigma_j)$; the direct second-derivative expectation vanishes. Inverting the triplet operator yields the displayed positive denominator. This is a Brillouin-zone integral at fixed spacing, not a continuum limit.

Put $a = c^2/U^2$ and $m_n = \binom{2n}{n}/4^n$. The positive geometric series for that denominator gives the exact rational lower sum

$$R_N = 1 + 16a \sum_{n=0}^{N} (4a)^n \sum_{i+j+k=n} \frac{n!}{i!j!k!}(m_i - m_{i+1}) m_{j+1} m_k, \qquad R = \frac{U\kappa}{c^2},$$

with $0 \le R - R_N \le 4a(12a)^{N+1}/(1-12a)$. The bound follows from $\sum \sin^2 k_l \le 3$ and $\langle \cos^2 k_x (\sin^2 k_y + \sin^2 k_z)\rangle = 1/2$. At the declared parameters $17/16 \le R \le 14/13$; the order-16 enclosure gives $R \simeq 1.06904778$ and kinetic coordinate mass $1/\kappa \simeq 59.86636$, whereas $U/c^2 = 64$. Under one clock/rod conversion $t_{\mathrm{phys}} = \tau t$, $x_{\mathrm{phys}} = \ell x$, the ratio $E\kappa/c^2$ is invariant. Such calibration cannot repair this particular same-speed relativistic mismatch. It does not exclude other scalings, reference states or candidate actions.

The energy split is also evaluated on ψ_K itself: $E_{\mathrm{int}} = UZ(K)$, $E_{\mathrm{kin}} = \langle \psi_K, T_K \psi_K\rangle$, and $\partial E/\partial U = Z(K)$ at fixed c, K. This is the response to the supplied action coupling, not a derivation of U from native repair maintenance or apparatus work.

7.2 The interacting taste boundary

Let Q retain a proper, exchange-symmetric momentum pocket in the $K = 0$ fiber, with $w = \langle d, Qd\rangle \in (0,1)$. The free action commutes with Q, while the contact projector gives exactly

$$\|[H, Q]\| = U\sqrt{w(1-w)}.$$

To see this, decompose $d = Qd + (1-Q)d$ into its two orthogonal normalized directions; the commutator has a two-dimensional off-diagonal block of that magnitude. On the $L = 6$ box the eight sign-of-cosine pockets have $w = 1/8$ each, contain the eight Weyl nodes, and are invariant under $k \mapsto -k$. A state $Qd/\sqrt{w}$ leaks into all seven omitted pockets under the same interaction. The squared short-time leakage coefficient is $7/64$, and the complete omitted-pocket probability is

about 0.26706 at model time two. Removing U removes this leakage. Dropping pockets is therefore not an exact interacting chiral restriction of this action. This excludes neither another regulator nor a controlled decoupling limit.

The same complete physical fiber admits a faithful finite Gibbs density, positive Euclidean reflection Gram matrices and the finite KMS identity for its declared even observables. These matrix facts do not establish a spatial-continuum reconstruction, physical vacuum or chiral gauge theory. The executable sources, complete records and proof boundaries are retained in the public `bound_pair`, `particle_clock`, `field_regions`, `field_stress` and `field_tastes` model packages.

7.3 The same pair's gauge current and contact residue

Couple the same hopping block to a supplied link phase, $h_{xy}(A) = e^{iA_{xy}} ic\sigma_a/2$. The complete CAR action obeys $H(A+d\alpha) = G(\alpha)H(A)G(\alpha)^*$ with $G = \exp(i\sum_x \alpha_x N_x)$. Its number current is $J_{x\to y} = \partial_{A_{xy}} H = i(B_{xy} - B^*_{xy})$, and $i[H, N_x] + \sum_y J_{x\to y} = 0$. These identities retain nonflat links, all tastes and the separated-pair space; the charge is CAR number, not an identified electromagnetic normalization.

Let P retain all onsite pairs, $Q = I - P$, $B = PTQ$ and $D = QTQ$, where $T = d\Gamma(h)$. For $E > \|D\|$, the exact contact operator is

$$F(E) = UI + B(E-D)^{-1}B^*, \qquad X(E) = (E-D)^{-1}B^*, \qquad M(E) = I + X(E)^*X(E).$$

A contact vector u with $F(E)u = Eu$ reconstructs the full eigenstate (u, Xu). Its norm and contact residue involve M, and its exact current is

$$\langle J\rangle = \frac{u^*(\partial_A F)u}{u^*Mu}, \qquad M = I - \partial_E F.$$

For a simple eigenvalue this also equals dE/dA. Omitting the metric gives an incorrect current by omitting the virtual separated-pair norm.

At leading order $H^{(2)}_{\text{pair}} = UI + BB^*/U$, two CAR hopping orders give $(BB^*)_{xy} = 2\det h_{xy}$ and hence $(H^{(2)}_{\text{pair}})_{xy} = c^2 e^{2iA_{xy}}/(2U)$. This derives charge-two pair transport on the same action. If $b \geq \|T\|, \|D\|$ and $C \geq \|BB^*\|$, its controlled remainder is

$$\|F(E) - H^{(2)}_{\text{pair}}\| \leq C\left(\left|\frac{1}{E} - \frac{1}{U}\right| + \frac{b}{E(E-b)}\right), \qquad E > b.$$

The complete-field response need not have the leading pair model's half-flux period. The source and independent current/residue checks in `model/gauged_field` retain that countercontrol. Link selection, physical clock and vacuum calibration, native protection and the chiral continuum remain separate requirements.

7.4 An exterior-matched native coherence and its limits

A separate construction tests localization directly on the native binary carrier. With endpoint weights $\theta = (25, 25, 8, \ldots, 8)$, its faithful reference is $\pi_{ij} = \theta_i\theta_j/7505$. Choose the actual seam $A = \{1, 7\}$. For each $k \notin A$, replace the reference block on $\{|1,k\rangle, |7,k\rangle\}$ by

$$\rho_k = \frac{\theta_k}{7505}\begin{pmatrix} 25 & 8 \\ 8 & 8 \end{pmatrix},$$

leaving every other entry unchanged. Each determinant is $136(\theta_k/7505)^2 > 0$; normalization and the diagonal are unchanged. The full exterior partial trace of $\rho - \pi$ vanishes. The global half trace distance is $D_0 = 776/7505$.

Every complete native record acts by a partial injection. Equal records preserve the signed difference of two load configurations, so a surviving $|1,k\rangle\langle 7,k|$ becomes $|1,l\rangle\langle 7,l|$. Its exterior partial trace is zero. Induction therefore proves equality of the joint exterior and entire native record tape for ρ and π at every finite history length. The preparation archive is distinct and is retained separately. This is a specified mixed state, not an arbitrary logical-qubit encoding. Its coherent core cannot move under these branches.

For the uniform all-seam scheduler, the ten spectator amplitudes evolve through an exact reversible substochastic matrix Q. With n the number of native attempts, their contrast obeys

$$D_n = \sum_k v_n(k), \qquad \sum_{n\geq 0} \frac{D_n}{D_0} = \frac{\theta^{\mathsf{T}}(I-Q)^{-1}\mathbf{1}}{\sum_{k\notin A}\theta_k}.$$

The integrated relative visibility is approximately 6.73354934 attempts for C, exactly 10/3 for E, and approximately 4.45400705 for the equal classical mixture. For E, $D_n/D_0 = (7/10)^n$. These are decays in native attempt count. They are not particle lifetimes in a calibrated physical time.

The reference flow supplies phase $\exp[-is\log(25/8)]$, where s is modular parameter. A supplied Ramsey analyzer followed by a complete native-current read has

$$\mathbb{E}j = -\frac{1649}{15010} + D_n\cos[s\log(25/8)].$$

Despite that phase, the same state satisfies $\mathrm{Tr}[(\rho-\pi)(-\log\pi)] = 0$ and $D(\rho\|\pi) = S(\pi) - S(\rho) > 0$. Clock sensitivity and an information deficit do not imply a nonzero energy or mass. Compiled rotations prepare all 66 mixture columns on the original 512 source addresses; the replay retains all 128 raw histories of a requested C-then-E continuation. The exact scheduler calculation separately covers all 30 seams.

Exact arbitrary-qubit correction has a stronger restriction. The existing full-78-state flagged-code theorem excludes double occupations and every native-edge pair from any exact code of logical dimension at least two, and makes each compressed occupation scalar. For any two-port core, exterior events whose every core extension is excluded have code probability zero but positive reference probability. The full core census therefore gives exterior half trace distance at least 96/395. This applies to arbitrary exact subspace codes, without assuming a sharp core occupation. For $N_H = n_0 + n_1$ on the permitted support, write $V^* N_H V = \nu I$. Up to scalar phase,

$$V^*\pi^{-is}V = [1-\nu+\nu e^{-is\log(25/8)}]I, \qquad p_{\mathrm{leak}} = 4\nu(1-\nu)\sin^2\!\big(s\log(25/8)/2\big).$$

An invariant exact code has $\nu = 0$ or 1 and a scalar logical clock. Approximate codes, other references and additional controlled services are outside this exclusion. The native construction supplies no protected boundary charge. Its source, exact rational checks and complete histories are retained in `model/native_localization`.

7.5 Moving the full band with a counted clock service

The native two-level clock cannot extend to the full Weyl/contact band by merely adding independent reference copies. Put $q = 25/8$. A single native reference has eigenvalues $(64/7505)q^m$, with $m = 0, 1, 2$ and multiplicities $45, 20, 1$. Every finite product has modular gaps in $\mathbb{Z}\log q$. The eta calibration fixes $\beta U = \log q$. At total momentum zero, the constant singlet obeys

$$\langle d, Td\rangle = \langle d, T^3 d\rangle = 0, \qquad \langle d, T^2 d\rangle = 6c^2 = 3U^2/32.$$

The compression to $d, Td/\|Td\|$ has matrix $U\left(\begin{smallmatrix} 1 & \sqrt{3/32} \\ \sqrt{3/32} & 0 \end{smallmatrix}\right)$. Its characteristic polynomial at $13/12$ is $-1/288$; min–max and $\|T\| < U/2$ give $13/12 < E_{\mathrm{pair}}/U < 3/2$. The eta dark state has energy zero. This noninteger difference excludes an invariant isometry intertwining both field levels with any finite independent native product at the same calibration. More generally, $\|(A - aI)V - VH/U\| \le \varepsilon$ for an integer-spectrum native generator A, any scalar a and isometry V implies $\varepsilon > 1/24$. Each of the two field energies must lie within ε of the shifted integer spectrum, whereas their separation is more than $1/12$ from any integer. This is an inherited-reference exclusion, not a prohibition on another reference or a counted simulation service.

Such a service is executed on the complete 8,128-dimensional sector using all 64 band momenta, scattering states and Weyl tastes. The additional reference premise is the globally normalized faithful density $\tau = e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$. It gives $\tau^{is} A \tau^{-is} = e^{-i\beta s H} A e^{i\beta s H}$ on the entire algebra. Separate normalization of momentum fibers fails on cross-fiber operators. The reference is supplied mathematically; its native selection and source preparation are not established.

The native record counts infer $\widehat{\beta}$, and a declared source-control service uses model duration $t = -\widehat{\beta} s$. Existing compiled local hopping/contact words and counted fermionic routes implement a symmetric 32-step split. All local occupation inputs are checked; the lifted count is 26,238,982 source pulses, rather than a literal replay of that many pulses on a global processor. The count acquisition separately uses 31,457,280 native attempts. A moving band packet is entangled with an untouched qubit, and all four terminal occupation-pointer outputs survive. For $|s| \le 1$, the confidence event gives $|\widehat{\beta} - \beta| \le 3/80$ and $|t| \le 5/4$. Thus the complete average half-diamond comparison obeys

$$\varepsilon \le \frac{9|s|}{256} + \frac{(7/4)^3 (5|s|/4)^3}{3\,32^2} + \frac{381169}{10^{12}} < 0.038565.$$

The bound is uniform over all sector inputs and finite untouched references; rejected count records give the same retained stop on both sides. No success probability divides the bound. It compares ideal completed source words; floating contraction residuals and hardware errors are separate diagnostics and implementation premises. Packet preparation, coherent loading and export, analogue control and the common field reference are declared inputs. The source-bound execution in `model/band_clock` therefore joins a moving finite band to a record-derived rate under an explicit added premise; it does not identify native repair with physical Hamiltonian motion.

Clock budget and energy offsets. The full-field enclosure $\|H - (35/96)I\| \le 77/96$ improves the rate-error term to $77|s|/2560$. With n symmetric split steps the bound becomes

$$\varepsilon_n \le \frac{77|s|}{2560} + \frac{(7/4)^3 (5|s|/4)^3}{3n^2} + \frac{381169}{10^{12}}.$$

At $|s| = 1$, $n = 21$ gives a bound below 0.037991; $n = 20$ exceeds the original 32-step ceiling. Thus 21 is minimal within this bound and ceiling, not among every possible simulation. The complete retained source schedule uses 17,219,334 pulses, saving 9,019,648. It retains all 8,128 states and four terminal outcomes. Acquisition, preparation and the global-reference premises are unchanged; hardware and floating errors are outside this ideal source bound.

The finite energy audit separates further identification questions. If the generator inside each momentum fiber is known up to a scalar, a connected tree of 63 admitted coherent comparisons would determine all 64 offsets up to one common scalar. These are mathematical access conditions, not constructed native detectors. Even such access cannot detect $H \mapsto H + cN$ within the fixed $N = 2$ carrier: that change is $2cI$ and leaves its normalized Gibbs state and dynamics unchanged. The eta-pair/dark contrast U remains an internal energy difference. Vacuum-relative rest energy

requires a further sector comparison or independently established preparation law; dispersion curvature concerns a different kinematic contract.

For the native localized coherence, complete raw histories preserve equality of its diagonals with the reference. If R_n and T_n are their respective pushforwards under the same complete instrument, then $\mathrm{Tr}[(R_n - T_n)(-\log T_n)] = 0$ on the image support at every finite history length. This uses no stationary state on the growing tape. It extends the zero-energy boundary to the full archive; positive relative entropy and decay in native attempt count do not furnish a mass.

Source-only conservation under classical feedback. For the faithful native H_{66} reference π and a conditional Heisenberg attempt map $\Phi(X) = \sum_b K_b^* X K_b$, completeness and reference invariance give

$$\mathrm{Re}\langle X, (I - \Phi)X\rangle_\pi = \tfrac{1}{2}\sum_b \|XK_b - K_bX\|_\pi^2, \qquad \langle X, Y\rangle_\pi = \mathrm{Tr}(\pi X^* Y).$$

Positive all-seam exploration and the complete branch algebra therefore make every fixed source observable scalar. Applied separately at each classical controller state, this excludes nonconstant source-only conservation on all admitted source/controller inputs; it does not assume a stationary growing archive. The exact weighted comparison gives $\Gamma \geq g_0/10240 > 9.3468\,10^{-7}$ per native attempt throughout the declared bounded family, where $g_0 = \beta_{20}/20$ uses the retained exact twenty-step population minorization. Hence $\|X - \mathrm{Tr}(\pi X)I\|_\pi \leq \|X - \Phi X\|_\pi/\Gamma$. Exogenous varying selectors sharing π also contract; correlated record-feedback mixing is not inferred. Joint relational charges, restricted preparations, active quantum recovery and the nonfaithful H_{78} stationary support are outside this source-only result.

8 The open joins

The joins below are conjunctive. Each must hold on the same continuing source process, and each has a test that stops its branch when it fails.

Native defect sector. Exhibit a state or observable sector of the unchanged law, including its quantum, complete-history and federated or refined extensions, with a boundary invariant preserved by every admitted branch; derive its coefficient group, show localization with reference-like response outside a bounded core, and distinguish autonomous protection from active recovery. A proof that every admitted-branch invariant on the declared carrier and refinement class is constant, as on the 78-state domain, removes the native defect mechanism for that law. A sector that exists only after freezing seams or supplying links and punctures belongs to a changed law and is tested as a new candidate.

Maintenance energy, kinetics and stress. Derive the excess energy of the defect over the reference from the same process, with a calibrated local clock, its kinetic terms and its stress, so that the same energy enters the modular and stress dictionary used by the gravitational layer [5]. A maintenance quantity that changes under admissible rescheduling while the observable response is unchanged is not an energy. A rest energy from maintenance that differs from the rest energy read from dispersion with the common signal speed rejects the maintenance reading of mass for that law.

Transport and identity. Show that moving the core transports its invariant, and derive the rules for pair creation, annihilation, fusion and decay from the same law. Obtain spin and statistics, for example from defect exchange, together with charges, currents and any family structure on the same defects. A charge that moves only through supplied link updates, or labels that are attached without a transport derivation, fail this join.

Interacting chiral description. Establish locality, a state and vacuum, the chiral measure and anomaly control, dynamical gauge interactions, and controlled volume and continuum limits, with massless radiation and gauge response arising as transport or collective excitations of the same construction. Extra regulator species, dark modes and complementary modes stay in the census. A proof that the admitted native regulator class keeps balanced chiralities in every refinement limit excludes the chiral spectrum for that class.

Frozen dictionary and matched spectrum. Freeze the source, observation map, calibration and uncertainties before the comparison, retaining a hash, date, decision rule, all tested alternatives and prior target exposure. The charged-spectrum benchmark is nine running charged Yukawas and the four independent CKM parameters in compatible conventions; neutrino masses and their mechanism are a further obligation. Compare ratios within a stated scheme and scale, then sector strengths, mixing and the operational absolute scale. A source-only GeV value is equivalent to a source-only dimensionless clock ratio, because the conversion through the defining cesium frequency is exact in SI units. A failed frozen comparison rejects that law and dictionary; changing the rule creates a new recorded candidate, and registration after exposure to known masses makes a postdiction.

Fixed boundaries. There is no strictly positive global vacuum mass gap in the intended infinite-volume theory with a massless photon, so sector gaps and massive thresholds name their state, sector and limit; a positive repair relaxation gap or finite-cavity spacing is not a physical mass gap. Quark masses require renormalization, matching and confinement observables, and confined gluons are not observable free particles. Unstable masses require resonance conventions, a pole on a stated sheet or a controlled line shape [18]. Charged sectors with massless gauge fields require an infrared treatment instead of an isolated one-particle pole [19, 20]. Under the hypothesis, conjugate defect classes supply antiparticles; a matter asymmetry additionally requires a physical CP-odd current, its state, generator, clock and thermal history.

Reproduction

The source operators, invariant projectors, Higgs tensors, native decision experiment, corrected-code control, record-memory protocol and hierarchy diagnosis are executable in the public repository with the observer construction [7, 1]. Their receipts distinguish exact algebra, floating-point diagnostics and empirical comparisons. A hash check verifies custody; it does not replace execution or extend a theorem's physical scope.

A Finite flavor constraints from a settlement rule

The constraints below concern the icosahedral carrier and a strict-descent settlement rule. That rule differs from the native seam rule of the executable observer, which sets the endpoint loads to the floor and ceiling of their mean and assigns an odd remainder by a fair coin; at load difference

one the native rule waits or swaps, so its balanced class keeps moving [7, 8]. The results are exact statements about the settlement rule and about the carrier geometry.

Quadrupole carrier. Let b_i be the centered totals of the record counts at the two ports of antipodal pair i, so that $b \in W_5 = \{b \in \mathbb{R}^6 : \sum_i b_i = 0\}$, and set $T(b) = \sum_i b_i(n_i n_i^{\mathsf{T}} - I_3/3)$ for unit axis vectors n_i. The icosahedral inner products give $T^*T = (4/5)P_5$ with $P_5 = I_6 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/6$, so W_5 is the geometric quadrupole carrier, irreducible under A_5. A supplied face axis has normalizer D_3, with $W_5|_{D_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2}$, whereas the physical geometric triplet restricts as sign $\oplus$ $\mathbf{2}$. The full native 30-seam port Laplacian acts on W_5 as $6I_5$, so a D_3 jump process on W_5 is a further dynamical hypothesis. With $\mathrm{Tr}(T_a T_b) = \delta_{ab}/2$, the color observable $A = (\sum_a T_a^2) \otimes P_0$ on $\mathbb{C}^3 \otimes W_5$, for the canonical rank-one face-axis projector P_0, has $\mathrm{Tr}\, A/15 = 4/15$ and $\mathrm{Tr}\, A^2/15 = 16/45$. Its identification as a mass response is a physical hypothesis.

Settled endpoints. The settlement rule transfers one record across a seam only if $N_i - N_j \geq 2$; it preserves $\sum_i N_i$ and strictly lowers $\sum_i N_i^2$, so every bounded fiber is finite and forward invariant. In the sector $N_i \in \{0, 1, 2\}$, $\sum_i N_i = 12$, repair schedules are orderings of maximal matchings of the active bipartite graph; exhaustive enumeration gives 73,789 states and 303 settled states in nine rotational orbits, with largest minimum cost four and longest schedule six. Settledness implies $|N_i - N_j| \leq d(i, j)$, and at totals divisible by twelve the 303 settled states exhaust all signed-integer endpoints. Their quadrupole spectra are $\{0, 0, 0\}$, $\{-2/\sqrt{5}, 0, 2/\sqrt{5}\}$, $\{-1-1/\sqrt{5}, 1-1/\sqrt{5}, 2/\sqrt{5}\}$ and its negative, so under an affine log-mass readout a nondegenerate ordered spectrum has adjacent log-gap ratio in $\{1, R, R^{-1}\}$, $R = (3/\sqrt{5}-1)/2$. At arbitrary total, the settled states reduce to 6,077 classes modulo uniform shift with exactly 31 quadrupole spectra, every integer settled state obeys $\mathrm{Tr}\, Q^2 \leq 24/5$, and the shape invariant $\chi = (\mathrm{Tr}\, Q^3)^2/(\mathrm{Tr}\, Q^2)^3$ takes eleven exact values, giving nineteen admissible log-gap ratios. These restrictions select no physical total, endpoint, scale or attachment.

A driven record state. An additional renewal protocol appends one of the twenty-four signed port events uniformly after each complete settlement and settles again with fresh uniform proposals. On the 6,077 settled relative configurations it defines an irreducible kernel of period two with a unique positive stationary law, $\mathbb{E}_\pi Q = 0$, all odd centered moments zero, and $\mathbb{E}_\pi\, \mathrm{tr}\, Q^2 \simeq 1.4647638612$. Stationarity of the bounded cost $W(N) = \sum_i (N_i - S/12)^2$, which rises by $11/12$ per preparation on average and falls by two per accepted repair, gives exactly $11/24$ accepted repairs per preparation. The process is not reversible; it is a driven stationary record law, not a thermal equilibrium or quantum vacuum, and its moments transfer to a physical response only through an identification.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 flavor potentials and residual-stabilizer selection [40, 59].
- Burnside_orbit_counting [42, 32].
- CAR_exterior_square_representation [31].
- Classical electroweak symmetry breaking and Goldstone counting in the Standard Model Higgs sector [95, 82, 49].

- Classical Koide relation, standard one-loop running and the existing OPH certified alpha and tau artifacts [60, 10, 82].
- complex_pole_to_running_width_Breit_Wigner_conversion [26, 88].
- cumulant expansions [61].
- De Broglie internal-clock and Zitterbewegung phase kinematics [38, 51, 86].
- Electroweak phenomenology, exterior weak multiplicities, hierarchy-problem literature, and commuting-square naturality diagnostics [16, 25, 90, 91, 67].
- exact_quadratic_field_linear_algebra [34].
- exact_rational_linear_algebra [92, 27].
- Exterior or Fock algebra and Standard Model exterior embeddings [25, 31, 45].
- exterior_algebra_and_CAR [31, 55].
- Finite clock models, Hamiltonian scaling, and determinant-line mass invariants [81, 52].
- Finite-group heat kernels [39, 85].
- finite_absorbing_Markov_chains_and_triangular_spectra [57, 52].
- finite_absorbing_Markov_resolvent [77].
- finite_dimensional_min_max_and_rank_one_resolvent [28, 56].
- finite_Gibbs_modular_flow [31].
- finite_group_invariant_operator_averaging [87].
- finite_group_representation_theory_and_Schur_lemma [87, 44].
- finite_irreducible_periodic_Markov_chain_stationarity [62, 77].
- finite_quantum_error_correction_effect_criterion [94].
- finite_random_unitary_channels_and_class_generators [94, 39].
- Flavor-basis invariants, singular-value decompositions, and parameter non-identifiability [54, 29, 83].
- Free plane-wave evolution and de Broglie phase kinematics under translation flows [38, 96].
- Georgi–Jarlskog/Clebsch flavor textures, one-loop Yukawa running, and lattice light-quark ratios [46, 66, 14].
- Gibbs_relative_entropy_identity_and_finite_KMS [31].
- Harremoes and Topsoe (2001), DOI 10.1109/18.959272, Shannon entropy at fixed index of coincidence [12].
- Hellmann_Feynman_identity_and_spectral_perturbation [56].
- Koide's charged-lepton relation, Hermitian C3 circulants, and finite GNS constructions [60, 36, 30].
- Kraus_complete_positivity_and_trace_distance_contraction [94].
- maximal_matching_combinatorics [63].
- moment_and_cumulant_algebra [74, 80].
- numerical_quadrature_and_coupled_ODE_shooting [89, 37].

- Pauli algebra sigma_a sigma_b = delta_ab + i epsilon_abc sigma_c and the Pauli Hamiltonian derivation of g = 2 [79, 84].
- PDG and FLAG quark summaries [16, 14].
- perturbative renormalization-group transport [65, 64].
- positive_Hermitian_kinetic_normalization [52, 58].
- Primary mathematical and physical sources [25, 76].
- QCD running-to-pole conversion [33, 75].
- repulsively_bound_lattice_pairs [97].
- Residual-symmetry flavor mixing and icosahedral axis geometry [59, 40].
- scalar_ODE_parameter_sensitivity_and_comparison_barriers [50, 93].
- Schur_complement_differentiation [99, 68].
- Signed graph Laplacians, integer positive-definite matrices, and spectral residual estimates [98, 53, 78].
- SMDR_1_3_complex_pole_and_local_charged_current_matching [73, 72].
- Standard Model Higgs near-criticality calculations [43, 17].
- Standard_Model_beta_functions_top_only_specialization [66, 64].
- Standard_Model_two_loop_beta_functions_and_effective_potential [64, 41, 69].
- standard_SU3_fundamental_Casimir [47, 82].
- strict_convexity_and_exponential_family_KL_projection [35, 24].
- SU3_and_SU2_intertwiner_identities [47, 44].
- Tree-level charged-current couplings g/sqrt(2) with weak-isospin raising and lowering operators [95, 82].
- Tree-level electroweak neutral-current couplings Q = T3 + Y and T3 - Q sin^2 theta [48, 82, 95].
- TSIL_and_3VIL_loop_integral_evaluation [70, 71].

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